

Multi-objective optimisation for constructing cyclic appointment schedules for elective and urgent patients

T. Meersman · B. Maenhout

Abstract In this paper, we study the construction of a cyclic appointment schedule in an outpatient department. In particular, we determine the capacity distribution between elective and urgent patients and the scheduling of the time slots reserved for these patients such that the operational waiting times of elective and urgent patients are minimised. The proposed solution methodology devises a Pareto set of cyclic appointment schedules based on these waiting times with different capacity allocations for urgent patients. An approximation of the Pareto set of non-dominated schedules is obtained using a multi-objective archived simulated annealing heuristic. To accurately validate the cyclic appointment schedules, we incorporate operational decision-making via scheduling individual patients. To this end, we simulate operational variability, i.e., patient arrivals, no-show behaviour, punctuality and scan durations, based on real-life input data. The patients are assigned one-by-one using an online scheduling rule. Computational experiments are conducted with a real-life case study. We compare different appointment scheduling rules and discuss the impact of the capacity distribution between elective and urgent patients and the timing of urgent slots in the cyclic appointment schedules. The results show that the distribution of capacity between patient types, the timing of urgent slots and appointment rules all have significant impacts on patient waiting times. Appointment waiting times improve when urgent slots are spread equally over and throughout the days considered and when the Bailey-Welch rule is used to schedule patients. Trade-offs between elective and urgent waiting times resulting from different capacity distributions or slot timing are exemplified via a Pareto front. The proposed method outperforms relevant single-pass methodologies, and we demonstrate that its performance is strengthened thanks to the integrated optimisation of strategic, tactical and operational decisions.

Keywords: Cyclic appointment scheduling; Outpatient department; Urgent patients; Multi-objective simulated annealing; Simulation

1 Introduction

In recent years, healthcare providers have been put under a great deal of pressure to provide high-quality care in a cost-efficient manner. In addition to clinical competences, perceived quality of service and patient satisfaction are increasingly reflected in patient waiting times (Huang, 1994; Anderson et al., 2007). In an effort to reduce costs and organise healthcare resources more efficiently, hospitals resort

T. Meersman · B. Maenhout
the Faculty of Economics and Business Administration, Ghent University, Tweeckerkenstraat 2, 9000 Gent, Belgium
Tel.: +3209-264-98-32
E-mail: tine.meersman@ugent.be

to using outpatient services. An outpatient clinic is the part of a hospital designed for the provision of diagnoses or treatments of patients that do not require a bed or an overnight stay (Zonderland, 2014). A well-designed appointment system can help improve the utilisation of expensive personnel and equipment-based medical resources and reduce waiting times for patients (Cayirli & Veral, 2003). Decisions related to the design of an appointment system are typically decomposed into three hierarchical phases because of the complexity of such systems and the different relevant time horizons during the different phases (Cardoen et al., 2010; Hulshof et al., 2012). The *strategic phase* addresses structural patient mixes and capacity decision-making over a long-term planning horizon and is based on highly aggregated information and forecasts (Fügener et al., 2014). During this phase, resource capacity decisions related to staff, equipment and space are made. In the *tactical phase*, the use of this capacity is determined by addressing the organisation of the operations/execution of the relevant healthcare delivery process. In this phase, capacity is assigned over time to specific patient types. The *operational phase* involves short-term decision-making. Based on the strategic and tactical decisions previously made, execution plans are designed at the individual patient and resource level. The main task during this phase involves scheduling or sequencing individual patients and coping with the arisen operational variability.

In this paper, we study the appointment system of a Magnetic Resonance (MR) Imaging department from a real-life case study involving a hospital that is located in Belgium. This hospital recognises that their urgent patients have to wait too long for an MR scan, as the capacity allocated to urgent patients may be insufficient and the reserved time slots for urgent patients are concentrated in the afternoon blocks. Capacity and time-slot allocation problems are recurring problems in most hospitals in Belgium, which impact patient satisfaction. The goal of this study is to minimise the appointment waiting times of elective and urgent patients while considering impacts on scan waiting times and daily overtime. To this end, we aim to redesign the outpatient appointment schedule consisting of the time slots dedicated to only elective and urgent patients. The developed appointment schedule is cyclic, implying that the schedule repeats after a certain sequence of days or a so-called period to meet appointment requests, which are assumed to adhere to a cyclic pattern (Kortbeek et al., 2014). From this perspective, we consider both strategic and tactical appointment scheduling decisions related to the capacity distribution between elective and urgent patient types and the timing of these time slots over the planning horizon, respectively. These decisions impact the waiting times of both elective and urgent patients. Including too many urgent slots leads to fewer time slots for treating elective cases and, hence, to longer elective appointment waiting times and vice versa. To accurately determine the required cyclic appointment schedule, operational decisions related to the scheduling of individual patients are included in our approach. As a result, operational variability, i.e., patient arrivals, scan durations, punctuality and no-show behaviour, is simulated, and different online appointment rules for scheduling individual elective patients are considered. These sources of uncertainty are simulated based on historical data retrieved from the real-life case study. All this information enables us to monitor the actual execution of an appointment schedule and to identify the best interrelated decisions regarding capacity allocation between patient types, the allocation of time slots throughout the cyclic horizon and appointment scheduling rules.

The solution methodology splits the problem in different sub-problems, and appointment scheduling rules are established for each one. For each appointment scheduling rule, we try to find the Pareto set of optimal solutions based on the appointment waiting times of elective and urgent patients such that decision makers can make focused trade-offs. The approximation of the Pareto set of non-dominated cyclic slot schedules is obtained using a multi-objective archived simulated annealing (SA) heuristic. Since uncertainty is incorporated in the simulation, the Pareto front is formed based on estimated averages. The model construction and data are based on a real-life case study. However, in order to generalise our findings, we perform an adequate sensitivity analysis of the case-specific conditions, capacity distribution, timing of urgent slots and appointment scheduling rule. The results reveal a trade-off between the

waiting times of elective patients and urgent patients. To minimise the waiting times of urgent patients, the slots should be distributed evenly over the planning horizon and throughout the examined days to some extent. The Bailey-Welch rule should be applied to determine the appointment times of elective patients.

The remainder of this paper is organised as follows. In Section 2, we discuss the relevant literature. In Section 3, we give a precise problem description and a mathematical formulation of the problem under study. Section 4 discusses the multi-objective optimisation methodology used. In Section 5, we validate the performance of the proposed procedure and explore different settings of the parameters characterising the cyclic appointment schedule. Section 6 provides concluding remarks and directions for future research.

2 Related literature

Appointment scheduling in an outpatient department involves the assignment of patients to suitable time slots for future appointments. For an exhaustive review of the literature, we refer to Ahmadi-Javid et al. (2017), Cayirli & Veral (2003) and Gupta & Denton (2008). Most of these studies focus on only the scheduling of elective patients and investigate the timing of appointments and patient acceptance/rejection (Robinson & Chen, 2010; Peng et al., 2014). Different objectives, such as minimising resource costs (e.g., overtime), minimising time-based measures (e.g., waiting/flow times), maximising resource capacity utilisation or optimising fairness and congestion measures, have been investigated in different forms. The objectives considered in this paper, i.e., minimising access time (or the time between patient requests and appointment times) and the waiting times of non-elective patients, have been researched in only a minority of the papers. Several papers (e.g., Gerchak et al. (1996); Sier et al. (1997); Blake et al. (2002); Lamiri et al. (2008); Van Huele & Vanhoucke (2014)) consider multiple objectives and formulate a single weighted objective function. However, Drexel & Nikulin (2008) state that it is very challenging to select weights for a multi-objective optimisation problem and that these weights are highly subjective. In this paper, however, we avoid the definition of a priori weights and construct a Pareto front of solutions, which enables an a posteriori focused selection by decision makers between a limited set of schedules where there is a trade-off between both objectives.

In the literature, several online and offline algorithms have been developed to schedule elective patients. Offline algorithms assume that the set of patients is known and is scheduled once in a static setting (Qu et al., 2007; Van Huele & Vanhoucke, 2014). Online rules operate in a dynamic environment, and patients are scheduled one-by-one in real time as they request appointments. The simplest rule is the first-come-first-serve (FCFS) rule. The FCFS rule typically divides a service session into equal slots of a length equivalent to the mean service time. Patients are scheduled in the chronological order in which they asked for appointments. A second well-known rule in appointment scheduling is the Bailey-Welch rule (Bailey, 1952; Welch & Bailey, 1952). According to this rule, patients are scheduled on a FCFS basis, but the first two patients are assigned to the first time slot. The subsequent patients are given appointment times that are earlier than their postulated time slots. Hence, this rule involves an overbooking of the first time slot to deal with the uncertainty of actual service times and no-shows. According to the research of Kaandorp & Koole (2007), the Bailey-Welch rule may lead to long waiting times. The individual-block/fixed-interval rule, also called the benchmark rule, assigns patients appointment times equal to the start times of the slot before theirs (Cayirli et al., 2006). This rule is generalised to an individual-block/variable-interval rule where for every patient an appointment time is determined that is earlier, equal or later than their actual slot time. Other rules, namely, multiple-block rules, assign multiple patients to one block. According to the two-block/fixed-interval rule, service times are divided into blocks equal to twice the mean service time. Two patients are assigned to each block and both

these patients get an appointment time equal to the start time of the block (Bard et al., 2016). Many studies compare these rules and several variants thereof to identify the best scheduling rule under different parameter settings, e.g., the probability of no-shows, punctuality, the coefficient of variation of service times and the number of customers per service session (Ho & Lau, 1999; Cayirli et al., 2006; Creemers et al., 2020). Their general conclusion is that rules designed to minimise customer waiting times perform best in most environments. However, they agree that the selection of a rule remains highly dependent on the environmental setting. Creemers et al. (2020) find that a variant of the benchmark rule, the dome rule, performs best when focusing on the waiting times of elective patients. As this rule hardly under-performs in terms of other criteria, they conclude that the dome rule is superior across all environments. In addition, a variant of the individual-block/variable-interval rule proposed by Ho & Lau (1999) performs very well, independently of the number of patients per session and the coefficient of variance relative to service times. This rule gives customers appointment times that are earlier than their preassigned time slots depending upon the average service time and the standard deviation. Note that these online rules have been devised and evaluated only for the scheduling of elective patients, while their impact on the waiting times of (non-elective) urgent patients remains unclear. In this study, online scheduling is applied to evaluate accurately operational performance; moreover, the best performing rule is determined in view of the waiting times of both elective and urgent patients.

In most studies in literature, only elective patients are considered and a certain reserve capacity for emergency and/or urgent patients is assumed. The arrival of the latter patients is then addressed via insertion using a reactive approach that involves rescheduling elective patients (Pham & Klinkert, 2008; Baretto et al., 2018). There are only a few studies in the domain of appointment scheduling that include strategic and tactical planning decisions related to the allocation of capacity to patient groups (Wiesche et al., 2017; Klassen & Rohleder, 1996; Zhou et al., 2018) and determine the timing of time slots for non-elective patients, i.e., walk-in, urgent and/or emergency patients (van der Lans et al., 2006; Wu et al., 2012; Kortbeek et al., 2014; Wu et al., 2014; Cayirli & Yang, 2014; Morikawa & Takahashi, 2017), which is the subject of this study. The lives of emergency patients are in danger, while those of urgent and walk-in patients are not. Hence, when minimising waiting times, decision makers prioritise the waiting times of emergency patients above those of urgent and walk-in patients, and emergency patients are typically inserted as soon as possible between elective patients. The arrival of non-elective patients in the context of outpatient scheduling is only studied to a limited extent. Rising et al. (1973) try to smooth the number of elective appointments over individual weeks and days to complement the number of non-elective patients and find a good schedule via trial-and-error using simulation. Klassen & Rohleder (1996) incorporate tactical decisions in appointment scheduling. Given the number of slots reserved for non-elective patients, they evaluate several timings of slots to treat urgent patients in combination with a number of different scheduling rules for elective patients. They find a low correlation between the appointment rules tested and the timing of the slots reserved for urgent patients. The individual-block/variable-interval schedule in which the first five patients are scheduled earlier than their actual slot times, performs very well in terms of minimising idle time but leads to increased waiting times. When the time slots allotted to non-elective patients are scheduled earlier in the day, the waiting times of patients decrease but the idle time of the service system increases. An important assumption of the aforementioned study is that the number of time slots for urgent patients is fixed. In this paper, we extend the study of Klassen & Rohleder (1996) by relaxing this assumption. More precisely, we embed strategic decisions by considering the number of time slots for urgent patients as a decision variable in the optimisation problem under study. Wiesche et al. (2017) studied appointment scheduling in the context of primary care, where walk-in patients are common. They tried to minimise waiting times and maximise capacity utilisation by finding the optimal capacity distribution between elective patients and walk-in patients. This is accomplished by reserving a minimal number of time slots for elective

appointments such that the time slots available for walk-in patients are maximised. However, the authors remark that there should be enough time slots for elective patients since pushing appointments further in the future increases the probability of no-shows. Appropriate time slot scheduling for elective patients helps level the workload, i.e., the number of patient visits, by shifting elective patients to days when fewer walk-ins are expected. Morikawa & Takahashi (2017) propose that time slots should be distributed into different blocks and that walk-in patients should be directed to time slots situated at the end of each block. Wang et al. (2019) use both overbooking and leaving appointment slots open to tackle the uncertainty of no-shows and walk-ins. Though these authors include time-dependent no-show behaviour, they assume that elective patients are punctual. Their objective is a cost function of waiting time per patient type, idle time and overtime. They find that when no-show rates are higher and the walk-in volume is lower, the optimal schedules show a high degree of overbooking during the earlier slots of sessions. In this paper, we integrate strategic, tactical and operational decision-making to determine the capacity and timing of dedicated time slots reserved for urgent patients while minimising the access times of both elective patients and urgent patients instead of walk-in patients, via application of a multi-objective optimisation procedure. In contrast to traditional approaches, neither the capacity nor the timing of the slots reserved for urgent patients are defined in advance or regulated via additional constraints or heuristic rules-of-thumb. The proposed methodology determines the capacity and timing of these slots internally based on operational performance to better align cyclic appointment schedules with stochastic patient arrivals, no-shows, punctuality and service times.

van der Lans et al. (2006) optimise the timing of time slots reserved for emergency patients, called break-in-moments (BIMs), in an operating room department. The main idea is to schedule elective cases in a way that distributes BIMs equally over individual days to minimise the waiting times of emergency patients. Their approach employs a two-stage heuristic to determine the timing of emergency time slots. In the first stage, the shortest processing time heuristic is used to minimise the time intervals between consecutive BIMs. In the second stage, the slot schedule is further improved via a simulated annealing approach. Many other studies have employed the idea of creating BIMs (e.g., van Oostrum et al. (2008); Davenport et al. (2014)). Wu et al. (2012) developed a local search technique that swaps the time slots of two patient assignments and guarantees that an optimal solution is found.

Appointment scheduling problems have been typically investigated via analytical methods (e.g., queuing theory and mathematical programming) or simulation techniques (Cayirli & Veral, 2003; Ahmadi-Javid et al., 2017; Jin et al., 2011). The drawback of analytical methods is that they face difficulties in dealing with complex environments. Simulations are suitable for evaluations of the performance of alternative appointment systems and for understanding the relationships between various environmental factors and various performance parameters. However, simulation techniques cannot be used to search for an optimal solution, as all possible solutions must be predetermined in advance. Recent approaches make use of simulation-based optimisation, which combines simulation with meta-heuristic techniques to guide searches towards promising solutions (Law & Kelton, 2007). Simulation-based optimisation approaches rely on an optimisation procedure to identify the values of decision variables, which serve as input for the simulation program. The simulation program assesses the quality of the solution under uncertainty and returns optimisation objective function values. A simulation-based approach iterates between the optimisation procedure and the simulation program until a satisfactory solution is produced or the process terminates due to prescribed conditions. In their literature reviews, Amaran et al. (2014), Alrabghi & Tiwari (2015) and de Sousa Junior et al. (2019) indicate that meta-heuristics such as genetic algorithms, SA and tabu search are frequently used optimisation methods in the framework of a simulation-based optimisation approach. In recent years, the use of these optimisation techniques to improve the organisation of outpatient departments has become more widespread (e.g., Saremi et al. (2013); Rezaeiahari & Khasawneh (2020); Li et al. (2021)). In these papers, optimisation searches and

the acceptance of new schedules are determined by the output of a simulation model, which evaluates new schedules and generates a single objective value based on the weighting of the considered objectives. In this paper, we propose the adoption of a multi-objective simulated annealing approach to construct a Pareto front of cyclic appointment schedules. The algorithm relies on a schedule archive and exploits different dedicated operators to optimise the capacity reserved for elective and urgent patients and the timing of the associated time slots. The schedule performance, which is indicated by multiple objective values, is evaluated using discrete-event simulation to guide the optimisation search process.

3 Problem Definition and Formulation

In the problem under study, we construct a cyclic appointment schedule for an outpatient department with a single server, taking operational variability into account. Strategic and tactical decisions are integrated with operational decision-making as follows:

- *Strategic and tactical decisions* correspond to the capacity distribution between different patient types and the position of their time slots over the cyclic planning horizon, respectively. The available capacity is divided between two patient types, i.e., elective patients (*el*) and urgent patients (*ur*). As the scan type of an elective patient is known in advance and the expected duration may differ across scan types, the elective slots could be divided further into multiple types. This leads to specific sets of slot types, i.e., urgent slots, elective slots for scan type *A*, elective slots for scan type *B*, etc. The capacity is allocated to certain time slots reserved for the related patient and scan type. These slots form a multi-day cyclic slot schedule that is repeated over time. Each day, which is also called a session, is divided into time slots during which one patient is served at a time. The length of each slot type is dependent on the associated patient and scan type. This slot assignment scheduling problem assigns various patient and scan types to time slots and reveals not only when a given type can receive an appointment but also the distribution of capacity between the different types. We assume that all the necessary resources, such as nurses and doctors, are available.
- The constructed cyclic appointment schedule is evaluated after *operational decisions* have been made and the operational schedule has been executed. The objective is to minimise the average appointment waiting times of elective and urgent patients. The appointment waiting time of a patient is equal to the time between his or her request for an appointment and his or her actual appointment. We also consider certain consequence variables, such as average daily overtime and average scan waiting time, as they are expected to be greatly influenced by strategic, tactical and operational decisions. To evaluate these objectives, which are situated on the operational level, a patient schedule is constructed based on a single-pass appointment rule and the proposed cyclic schedule. The operational variability resulting from human activity, i.e., patient arrivals, scan durations, no-show behaviour and punctuality of elective patients, is simulated in order to accurately evaluate patient waiting times.

In the following sections, we describe the problem in more detail. In Section 3.1, we present a mathematical model for an integrated problem that combines the distribution of capacity between patient and scan types, slot assignment and patient scheduling. Section 3.2 discusses operational decision-making, i.e., operational uncertainty, operational system dynamics and evaluation of operational schedules.

3.1 Integrated capacity distribution and scheduling

The integrated model under study defines the relevant objectives and constraints to construct a cyclic outpatient appointment schedule with a planning horizon of a single period. A period comprises multiple

consecutive days, and each day is composed of a set of time slots that are accessible to specific patient and scan types. The generated slot schedule is repeated over multiple periods in a cyclic manner so that individual patients can be scheduled. The proposed formulation applies the FCFS appointment rule to schedule individual patients. In the formulation, the following notation is used:

Sets

P	Set of periods in the planning horizon ($P = \{0, \dots, P - 1\}$, index p)
D	Set of days, also called sessions, in one period ($D = \{0, \dots, D - 1\}$, index d)
S_d	Set of available slots on day d ($S_d = \{0, \dots, S_d - 1\}$, where index s indicates the s^{th} slot of a particular day)
Q	Set of slot types ($Q = \{0, \dots, Q - 1\}$, index q)
I	Set of patients ($I = \{0, \dots, I - 1\}$, index i)

Parameters

b^1	Session start time
b^2	Session end time
f_{pd}^1	Session start time on day d in period p , i.e., $24 \times 60 \times (D \times p + d) + b^1$ (in minutes)
f_{pd}^2	Session end time on day d in period p , i.e., $24 \times 60 \times (D \times p + d) + b^2$ (in minutes)
l_q	Length of a type q time slot
σ_q	Standard deviation of the scan durations planned for slot type q
c_i^1	Call period of patient i
c_i^2	Call day of patient i
c_i^3	Call time of patient i on day c_i^2 (in minutes)
c_i	Call time of patient i , i.e., $24 \times 60 \times (D \times c_i^1 + c_i^2) + c_i^3$ (in minutes)
u_{kl}	1 if patient k has called before patient l , i.e., $c_k \leq c_l$, and 0 otherwise
r_{iq}	1 if patient i has to be assigned to a slot of type q and 0 otherwise
t_i	1 if patient i is an urgent patient and 0 if patient i is an elective patient
v_{kl}	1 if patients k and l are of the same type, i.e., $r_{kq} = r_{lq}$ ($\forall q \in Q$), and 0 otherwise

Decision variables

S_{ur}	Number of urgent slots per period
X_{dsq}	1 if time slot s on day d is a slot reserved for type q and 0 otherwise
Y_{pdsi}	1 if patient i is assigned to time slot s on day d during period p and 0 otherwise
Z_i	1 if patient i is scheduled and 0 otherwise
O_i	Start time of the slot assigned to patient i (in minutes)
A_i	Appointment time of patient i (in minutes)
W_i	Appointment waiting time of patient i (in minutes)
$E_{pdss'qi}$	1 if $Y_{pdsi} = 1$ and $X_{ds'q} = 1$ and 0 otherwise
N_{el}	Number of scheduled elective patients, i.e., $\sum_{i \in I} ((1 - t_i) \times Z_i)$
N_{ur}	Number of scheduled urgent patients, i.e., $\sum_{i \in I} (t_i \times Z_i)$

Mathematical formulation

$$\text{Minimise } \frac{1}{N_{el}} \sum_{i \in I} ((1 - t_i) \times Z_i \times W_i) + \sum_{i \in I} (M_1 \times (1 - t_i) \times (1 - Z_i)) \quad (1)$$

$$\text{Minimise } \frac{1}{N_{ur}} \sum_{i \in I} (t_i \times Z_i \times W_i) + \sum_{i \in I} (M_1 \times t_i \times (1 - Z_i)) \quad (2)$$

The considered objectives, eqs. (1) and (2), minimise the average appointment waiting times for elective and urgent patients, respectively, resulting from three interrelated decisions, namely, the number of slots per patient type, the timing of these slots and the selected appointment scheduling rule. The first two decisions are expressed via the decision variable X_{dsq} , which denotes the type of a certain slot. Set S_d comprises the set of available slots on a particular day, and these may be assigned to a specific

patient type via decision variable X_{dsq} . The decision variables X_{dsq} determine the set of slots planned for each day d and the number of urgent slots per period (S_{ur}). The last decision is expressed as a set of constraints and implemented via the decision variable Y_{pdsi} , which denotes the assignment of individual patients to slots. It is possible that a patient is not assigned to a time slot during the observed planning horizon. Such a situation may occur in case of capacity overload. Therefore, an extra decision variable, Z_i , is added to indicate whether a patient is scheduled. The large value M_1 ($\gg W_i$) ensures that the objectives maximise the number of patients scheduled.

Subject to

$$O_i = \sum_{p \in P} \sum_{d \in D} \sum_{s \in S_d} \left((f_{pd}^1 + \sum_{s' \in \{0, \dots, s-1\}} \sum_{q \in Q} (l_q \times X_{ds'q})) \times Y_{pdsi} \right) \quad \forall i \in I \quad (3)$$

$$A_i = O_i \quad \forall i \in I \quad (4)$$

$$W_i \geq A_i - c_i \quad \forall i \in I \quad (5)$$

$$\sum_{s \in S_d} \sum_{q \in Q} (l_q \times X_{dsq}) \leq b^2 - b^1 \quad \forall d \in D \quad (6)$$

$$\sum_{q \in Q} X_{dsq} \leq 1 \quad \forall d \in D, \forall s \in S_d \quad (7)$$

$$\sum_{p \in P} \sum_{d \in D} \sum_{s \in S_d} Y_{pdsi} = Z_i \quad \forall i \in I \quad (8)$$

$$\sum_{i \in I} Y_{pdsi} \leq 1 \quad \forall p \in P, \forall d \in D, \forall s \in S_d \quad (9)$$

$$\sum_{q \in Q} X_{dsq} \geq Y_{pdsi} \quad \forall p \in P, \forall d \in D, \forall s \in S_d, \forall i \in I \quad (10)$$

$$r_{iq} \geq Y_{pdsi} + X_{dsq} - 1 \quad \forall p \in P, \forall d \in D, \forall s \in S_d, \forall q \in Q, \forall i \in I \quad (11)$$

$$O_i + M_2 \times (1 - Z_i) \geq c_i \quad \forall i \in I \quad (12)$$

$$v_{kl} \times (O_k - O_l) \leq M_3 \times (2 - u_{kl} - Z_l) \quad \forall k, l \in I \text{ and } k \neq l \quad (13)$$

$$v_{kl} \times u_{kl} \times (Z_k - Z_l) \geq 0 \quad \forall k, l \in I \text{ and } k \neq l \quad (14)$$

$$\sum_{q \in Q} X_{d(s+1)q} \leq \sum_{q \in Q} X_{dsq} \quad \forall d \in D, \forall s \in S_d \setminus \{|S_d| - 1\} \quad (15)$$

$$X_{dsq} \in \{0, 1\} \quad \forall d \in D, \forall s \in S_d, \forall q \in Q$$

$$Y_{pdsi} \in \{0, 1\} \quad \forall p \in P, \forall d \in D, \forall s \in S_d, \forall i \in I$$

$$Z_i \in \{0, 1\} \quad \forall i \in I$$

$$O_i, A_i \geq 0 \text{ and integer} \quad \forall i \in I$$

$$W_i \geq 0 \quad \forall i \in I \quad (16)$$

The constraints of the integrated problem are given by the expressions (3) - (16). To determine the average waiting time of elective and urgent patients, the appointment waiting time of each scheduled patient is calculated. The relevant calculations of patient waiting times (W_i) are provided in equations (3) - (5). The waiting time of a patient depends on the time and day the patient calls (c_i) and on his or her assigned appointment time (A_i). In a simple FCFS appointment system, the latter is equal to the start time of the selected time slot (O_i). Equations (3) and (4) calculate the slot start time O_i and the appointment time A_i of each patient i in minutes. The right-hand side of equation (3) is non-zero only when patient i is assigned to slot s on day d , i.e., when $Y_{pdsi} = 1$. Equation (5) calculates the appointment waiting time W_i of patient i in minutes. Constraint (6) ensures that the aggregated times of the scheduled slots of a certain day are not greater than the session length. Constraint (7) ensures that each slot is characterised by one type q or is not assigned to any type. Constraints (8) to (11) involve the assignment of individual patients. Constraint (8) ensures that each patient is assigned to at

most one time slot. Constraint (9) stipulates that each time slot has no more than one patient assigned. Constraint (10) stipulates that a specific patient can be assigned to a slot if the time slot is dedicated to one of the patient types. Constraint (11) and constraint (10) jointly ensure that patients are only assigned to time slots reserved for their patient types, i.e., if patient i is scheduled ($Y_{pdsi} = 1$), the assigned slot type q of the respective time slot must be equal to the patient type q ($X_{dsq} = 1 \wedge r_{iq} = 1$). Constraint (12) ensures that patients are only assigned to time slots later than their call times. This constraint is only imposed for patients who are actually scheduled ($Z_i = 1$). If a patient has not been scheduled ($Z_i = 0$), a large value M_2 ($M_2 \gg 0$) is added to the left-hand side so that the constraint is redundant. Constraint (13) stipulates that if the two patients k and l are of the same type ($v_{kl} = 1$) and the call time of patient k is earlier than the call time of patient l ($u_{kl} = 1$), patient k should get an earlier appointment time than patient l , thus ensuring that patients are scheduled in the order of their arrival (FCFS). If the call time of patient k is later than the call time of patient l ($u_{kl} = 0$) or if patient l is not scheduled ($Z_l = 0$), a large value M_3 ($M_3 \gg 0$) is added to the right-hand side of the constraint so that the constraint is redundant. Constraint (14) requires that if patients k and l are of the same type and the call time of patient k is earlier than the call time of patient l , patient l can only be scheduled if patient k is also scheduled. Constraint (15) ensures that planned slots that are accessible to patients are successive, i.e., they follow each other directly in a consecutive manner, starting from the first time slot ($s = 0$). Constraint (16) formulates the domain of the variables. In defining a cyclic appointment schedule for a particular outpatient department, periodically reoccurring restrictions in the slot schedule may be added via the inclusion of additional constraints. Such constraints may imply the exclusion of a slot type for a particular time slot/day or after a particular time period has elapsed within a session.

Note that the proposed model is non-linear because of the product of the decision variables in, for example, equation (3). However, the model can be linearised if appropriate auxiliary decision variables are defined. In equation (3), in particular, we can replace the product between the variables $X_{ds'q}$ and Y_{pdsi} with the binary variable $E_{pdss'qi}$, which is equal to 1, if and only if both variables $X_{ds'q}$ and Y_{pdsi} are equal to 1. In this way, the nonlinear constraint (3) can be replaced with the equivalent set of linear expressions (17)-(21). Constraint (17) calculates the slot start time (O_i). Constraints (18) - (21) model the variables $E_{pdss'qi}$ by defining an AND relationship for the variables Y_{pdsi} and $X_{ds'q}$.

$$O_i = \sum_{p \in P} \sum_{d \in D} \sum_{s \in S_d} \left((f_{pd}^1 \times Y_{pdsi} + \sum_{s' \in \{0, \dots, s-1\}} \sum_{q \in Q} (l_q \times E_{pdss'qi})) \right) \quad \forall i \in I \quad (17)$$

$$E_{pdss'qi} \geq Y_{pdsi} + X_{ds'q} - 1 \quad \forall p \in P, \forall d \in D, \forall s \in S_d, \forall s' \in \{0, \dots, s-1\}, \forall q \in Q, \forall i \in I \quad (18)$$

$$E_{pdss'qi} \leq Y_{pdsi} \quad \forall p \in P, \forall d \in D, \forall s \in S_d, \forall s' \in \{0, \dots, s-1\}, \forall q \in Q, \forall i \in I \quad (19)$$

$$E_{pdss'qi} \leq X_{ds'q} \quad \forall p \in P, \forall d \in D, \forall s \in S_d, \forall s' \in \{0, \dots, s-1\}, \forall q \in Q, \forall i \in I \quad (20)$$

$$E_{pdss'qi} \in \{0, 1\} \quad \forall p \in P, \forall d \in D, \forall s \in S_d, \forall s' \in \{0, \dots, s-1\}, \forall q \in Q, \forall i \in I \quad (21)$$

As indicated, the proposed model above applies a simple FCFS appointment rule, which sets appointment times equal to the assumed time of service. Note that the presented model and corresponding formulas can be easily translated to other online scheduling rules for which patients are still assigned to slots according to a first-come-first-serve basis. However, they are given earlier or later appointment times that are different from the actual start times of their appointed slots, i.e., $A_i \neq O_i$. This is often done to cancel out possible variability in patient arrival time, treatment duration, etc. a priori. Notably, with such rules, calculations of the appointment time A_i for patient i based on equation (4) vary, which will be further explored in Section 4.

3.2 Operational decision-making

3.2.1 Operational uncertainty

The waiting times of elective and urgent patients are subject to operational variability. We assume that there are four types of variability related to patient arrival times, no-show behaviour, punctuality, and service durations. In the following, we describe the characteristics of operational variability for each type of uncertainty by means of a probability distribution with adequate parameters. The actual uncertainty in the data parameters is generated via a Monte Carlo simulation (cf. Section 4).

Uncertainty of patient arrivals - Operational variability impacts the arrival of elective and urgent patients and accordingly the associated waiting times of both types of patients. The variability in the arrival process of elective patients is related to their call times, which are typically generated based on a memoryless negative exponential distribution (Swartzman, 1970) dictating patient interarrival times. However, for each case, historical data should be used to determine which probability distribution best fits the arrival of calls, and appropriate parameter values should be estimated (e.g., Clavel et al. (2018); VanBerkel & Blake (2007); Guo et al. (2004)). The variability in the arrival of urgent patients is related to the moment these patients actually appear in the outpatient department. The interarrival times of urgent patients are typically generated based on a negative exponential distribution (e.g., Bargetto et al. (2018); Jin et al. (2011)).

No-show behaviour - No-show behaviour refers to the probability of elective patients not attending their appointments. This uncertainty does not influence appointment waiting times but highly impacts scan waiting times and daily overtime. We assume a time- and patient-independent no-show probability. The no-show probability distribution, i.e., the probability of a patient being a no-show, is dependent on the studied environment (e.g., a gamma distribution (Huang & Zuniga, 2012), Bernoulli distribution (Guo et al., 2004)).

Uncertainty of patient punctuality - The variability of patients' actual arrival times at the moment of appointment refers to the unpunctuality of patients, which instigates a difference between a patient's appointment time and his or her actual arrival time (Cayirli & Veral, 2003). Patients can arrive to their appointments early or late. Uniform (Bhattacharjee & Ray, 2016) and normal (Cayirli et al., 2006; Deceuninck et al., 2018) distributions are frequently used to model outpatient punctuality.

Uncertainty of service times - Operational variability may impact the actual duration of service encounters for patients. As a result, the appointment and scan waiting times of subsequent elective and urgent patients are impacted. In the literature, a wide range of service time distributions have been utilised depending on the specific case situation (e.g., triangular distributions (Lee & Yih, 2014), normal distributions (Johnson et al., 1997), and lognormal distributions (Rohleder & Klassen, 2002)).

3.2.2 Operational system dynamics

When calling for an appointment, a patient calls the front desk of the outpatient department, and the patient appointment time is set, i.e., the time the patient should arrive at the department following the appointment scheduling rule. This can be different from the start time of the slot appointed to the patient. These patients are given appointments in the order in which their calls arrive (FCFS), i.e., the next elective patient that calls the outpatient department is appointed to the next available time slot in the planning horizon that is dedicated to elective patients of that type. On the day of examination, these elective patients either actually arrive or do not show up. Elective patients can arrive late or early in comparison to their appointment times. In addition to elective patients, urgent patients without appointments may arrive at the department, and they are assigned to the next available time slot

dedicated to urgent patients. Given the urgency of these patients, they should be examined the same day. If no time slots are left on a particular day, these patients are assigned to overtime time slots, i.e., time slots that fall after the normal opening hours of the outpatient department, when all elective patients are served. Both types of patients are examined in the order set by the appointment schedule. We assume that it is known whether a patient is a no-show or a late arrival at the moment the next scheduled patient arrives. Therefore, the slot of a no-show patient is presumed to be idle. The earliest scan start time of the patient following a no-show patient is equal to the maximum of that patient's actual arrival and the time that the patient preceding the no-show patient leaves.

3.2.3 Evaluation of operational schedules

Cyclic slot schedules are evaluated based on their operational execution, relying on primary and secondary performance measures referring to the objectives and the consequence variables, respectively.

Primary performance measures: Objectives

Based on a list of patient arrivals and service times, patient schedules are realised and the related average appointment waiting times of elective patients (eq. (22)) and urgent patients (eq. (23)) are calculated. As these patient waiting times are subject to some operational randomness, these averages represent only point estimates of the true waiting times associated with the cyclic appointment schedule.

$$\overline{OV}_{el} = \frac{1}{N_{el}} \sum_{i \in I} ((1 - t_i) \times Z_i \times W_i) \quad (22)$$

$$\overline{OV}_{ur} = \frac{1}{N_{ur}} \sum_{i \in I} (t_i \times Z_i \times W_i) \quad (23)$$

The accuracy of the primary performance measure $\overline{OV}_m$ ($m \in \{el, ur\}$) can be determined by calculating the corresponding confidence interval CI_m based on the estimated standard error ($\hat{err}_m$). The estimated standard error is calculated as $\hat{err}_m = \hat{s}_m / \sqrt{N_m}$ and is thus determined by two factors, namely, the sample size (N_m) and the estimated standard deviation ($\hat{s}_m$). The estimated standard deviation $\hat{s}_m$ measures the variability between the appointment waiting times of N_m patients of type m and is calculated according to equation (24). The confidence interval is calculated according to equation (25). The smaller the confidence interval is, the more accurate the estimate is. α is a measure of desired accuracy and lies in the interval $[0,1]$. The t^* -value is the critical value of the confidence interval and depends on the accuracy level ($\frac{1+\alpha}{2}$) and on the degrees of freedom ($N_m - 1$).

$$\hat{s}_m = \sqrt{\frac{1}{N_m - 1} \times \sum_{i=1}^{N_m} (W_i - \overline{OV}_m)^2} \quad (24)$$

$$CI_m = \left[\overline{OV}_m - \hat{err}_m \times t_{\frac{1+\alpha}{2}, N_m - 1}^*, \overline{OV}_m + \hat{err}_m \times t_{\frac{1+\alpha}{2}, N_m - 1}^* \right] \quad (25)$$

Secondary performance measures: Consequence variables

Based on the effectuated patient schedule, we can evaluate the average scan waiting time and the average daily overtime. These variables are not explicitly taken into account in the objective function but could be interesting for tie breaking in case the decisions do not have a significant impact on the primary performance measures or simply to provide some extra information. The waiting time of patient i at the time of his or her appointment, which corresponds to the scan waiting time of this patient, can be

calculated based on equation (26). With the scan waiting times of individual patients, the average scan waiting time ($\bar{H}_{el}$) of elective patients and the related interval estimators can be calculated.

$$H_i = \max(\gamma_{i^*} + \delta_{i^*} - (A_i + \theta_i), 0)$$

with

i^*	The patient scheduled in the slot before patient i	(26)
H_i	The scan waiting time of patient i (in minutes)	
γ_i	The actual start time of patient i (in minutes)	
δ_i	The actual appointment duration of patient i (in minutes)	
θ_i	The punctuality of patient i (< 0 if too early, ≥ 0 otherwise) (in minutes)	

Average daily overtime ($\overline{OT}$) can be calculated by averaging the overtime OT_{pd} incurred over all the days of the considered periods. Overtime OT_{pd} equals the strictly positive deviation between the scan end time of the last patient i^* and the session end time (eq. (27)) of day d in period p . Based on daily overtime, the average daily overtime $\overline{OT}$ over the planning horizon and the related interval estimators can be calculated.

$$OT_{pd} = \max((\gamma_{i^*} + \delta_{i^*}) - f_{pd}^2, 0) \quad (27)$$

4 Solution Methodology

The integrated optimisation problem under study is characterised by many stochastic variables and an exponential number of possible scenarios, such that a stochastic optimisation approach is computationally intractable. In addition, the proposed model (1) - (16) is non-linear. Its linearisation requires the additional definition of numerous auxiliary (binary) variables and associated constraints, which hampers the performance of a (deterministic) method relying on the use of mathematical programming due to a large number of symmetries. To tackle the proposed appointment planning and scheduling problem, we combine optimisation and simulation techniques, i.e., simulation-based optimisation. Simulation-based optimisation is a technique mostly used for discrete-event stochastic systems where the size and the complexity of the focal problem makes it hard or even impossible to use classical optimisation methods (Gosavi, 2003). Figure 1 gives a general overview of the proposed solution methodology, which is explained in detail in this section.

In this study, decisions related to the planning and scheduling of patients are made on strategic, tactical and operational levels. Capacity decisions on the number of time slots reserved for elective and urgent patients are strategic decisions. The timing of these slots is a tactical decision. These decisions are optimised using a structured heuristic search technique, i.e., a multi-objective archived simulated annealing heuristic (Section 4.1). For each generated cyclic appointment schedule, a parametric simulation is effectuated to evaluate the quality of the obtained schedule on the operational level (Section 4.2). The simulation model mimics patient call times, no-show behaviour, punctuality, and scan durations and makes operational decisions, scheduling patients one-by-one based on an online scheduling rule. This simulation facilitates the evaluation of objectives and consequence variables. The goal is to identify the best strategic and tactical decisions such that appointment waiting times are minimised given an operational appointment rule. The multi-objective evaluation enables the approximation of a Pareto set of solutions such that decision makers can make focused trade-offs. The objective under study considers the average appointment waiting times of both elective and urgent patients. Assigning weights to solve multi-objective problems is highly subjective and not always possible (Drexel & Nikulin,

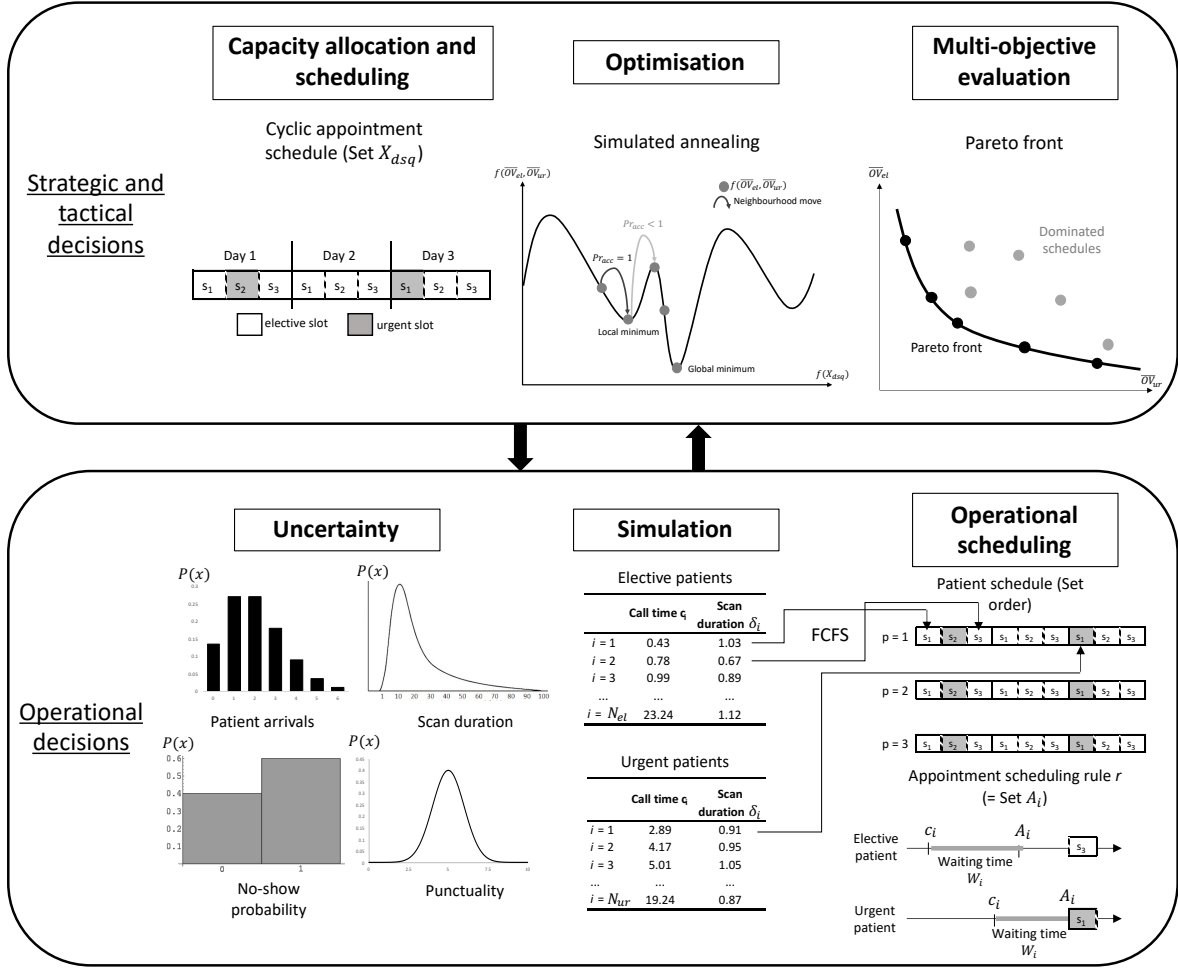


Fig. 1: A conceptual overview of the solution methodology

2008). Constructing a Pareto set of solutions resolves this issue and leaves the choice to decision makers (Czyzszak & Jaszkiwicz, 1998). A Pareto set is a set of solutions that are not dominated by any other solution in the sampled search space.

In summary, the SA algorithm feeds cyclic appointment schedules to the simulation model, which evaluates this input and returns the estimated objective values. These returned objective values form a link between the simulation and the optimisation algorithm, guiding the schedule update and archiving of retrieved Pareto schedules and hence directing the optimisation search process.

4.1 Multi-objective simulated annealing

The multi-objective simulated annealing algorithm generates a set of non-dominated schedules that converges to the global Pareto frontier as the temperature (T) decreases. The idea of constructing a Pareto set of solutions for a multi-objective optimisation problem in an SA framework instead of focusing only on an incumbent solution has already been explored (Czyzszak & Jaszkiwicz, 1998; Bandyopadhyay et al., 2008). A schedule is represented by an ordered list of 'slot' objects that each contain all the information about that slot (i.e., start time, duration, etc.). In the search for a Pareto front, a fixed number of temperature levels (N_{epoch}) are considered. At each temperature level, a fixed number of iterations (L) is carried out starting from an initial schedule. The procedure can be initialised by a completely random schedule or a good schedule. A good schedule could be either a real-life schedule

that is currently used in an outpatient department or a schedule found via another heuristic procedure such as a hill-climbing heuristic. This initial schedule is set as the current schedule (*current*) in the first iteration. In each iteration, a different cyclic appointment schedule is generated by employing a random neighbourhood operator to construct a new schedule (*new*) in the neighbourhood of the *current* schedule, i.e., the schedule that is retained from the previous iteration (Section 4.1.1). The newly generated schedule is compared with the *current* schedule and the schedules in the archive based on their objective function values. The replacement of the *current* schedule with the *new* schedule or the inclusion of the *new* schedule in the archive is determined by the probability of acceptance and the level of domination between the *new* schedule and the *current* schedule or the schedules in the archive (Section 4.1.2). The archive contains only solutions that are not dominated by the solutions considered so far in the search process and is updated when a newly constructed schedule dominates archived solutions. The probability of acceptance is further influenced by the temperature level, which starts at T_0 and decreases over time. The temperature is recalculated as $T = T \times \beta$ ($\beta \in [0, 1]$) after L iterations have been carried out at a specific temperature level. When the temperature decreases, the probability of acceptance decreases. This temperature increases the probability of convergence from the archive to the Pareto frontier and impacts the balance between diversification and intensification. Choosing a correct combination of β , N_{epoch} and L is important to increase the probability of convergence to the Pareto front. Algorithm 1 provides insight into the general components of the SA algorithm.

Algorithm 1 The multi-objective SA algorithm

```

1: Initialise SA parameters
2: Construct initial schedule
3: Set current = initial schedule
4: For  $counter_1 = 1 \rightarrow N_{epoch}$ 
5:   For  $counter_2 = 1 \rightarrow L$ 
6:     Construct new
7:     Update current and the archive
8:   Set Temperature  $T = T \times \beta$ 

```

4.1.1 The searched neighbourhood and move operator

A neighbouring solution of the *current* schedule is obtained by operating on the values of the decision variables X_{dsq} via either a change in the capacity distribution or a change in the slot sequence. These operations are conducted with a certain probability, i.e., Pr_1 or $1 - Pr_1$, respectively. In the case of a change in the slot sequence, two operations are available, i.e., an interchange operation or a shift operation. These operations are conducted with a certain probability, i.e., Pr_2 or $1 - Pr_2$, respectively. Algorithm 2 shows the steps to generate a neighbour of the *current* schedule according to the uniform random numbers u_1, u_2 and u_3 , which are generated in the interval $[0, 1]$. Note that some of these operations assume that a slot type with a longer duration is perfectly dividable by any slot type with a shorter duration. This is necessary to keep the daily capacity constant when slots of a different length are moved.

A new schedule is generated by one of the following operators:

- The *capacity* operator selects two different random slot types, q_1 and q_2 . Next, a random slot s_k of type q_1 is selected and changed to type q_2 , i.e., $X_{ds_k q_1} = 0$ and $X_{ds_k q_2} = 1$. If $l_{q_1} < l_{q_2}$, slots of type q_1 directly preceding or following s_k are removed until the length of the session is equal to the originally postulated session length. If $l_{q_1} > l_{q_2}$, the opposite operation is performed. Figure 2 illustrates all three possible capacity operations. If none of these are feasible, new slot types are selected.

Algorithm 2 Construct *new*

```

1: Set new = current
2: Generate  $u_1 \in U[0,1]$ 
3: If  $u_1 < Pr_1$ 
4:   Perform capacity operator on new
5: Else
6:   Generate  $u_2 \in U[0,1]$ 
7:   If  $u_2 < Pr_2$ 
8:     Perform interchange operator on new
9:   Else
10:    Generate  $u_3 \in U[0,1]$ 
11:    If  $u_3 < 0.5$ 
12:      Perform forward shift operator on new
13:    Else
14:      Perform backward shift operator on new

```

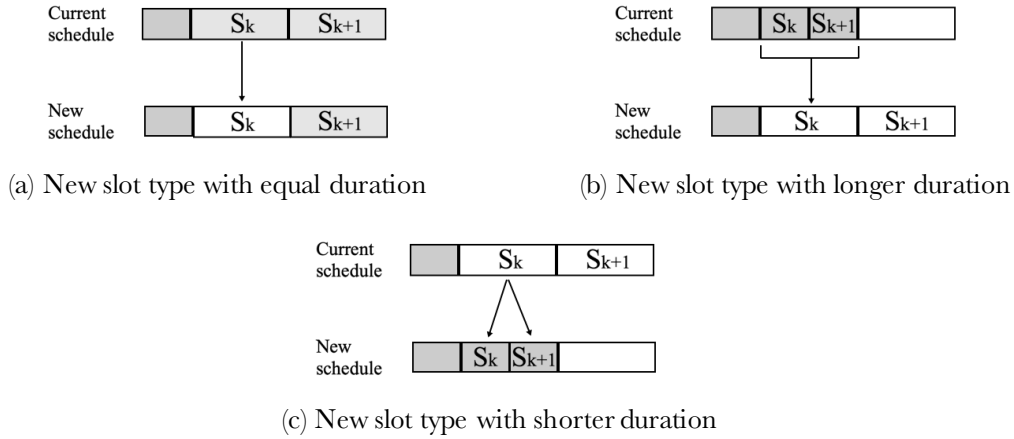


Fig. 2: Illustration of the capacity operator

- The *interchange* operator randomly selects two time slots, s_k and s_l , which are characterised by different slot types, q_1 and q_2 , respectively. The selected slots can either be on the same day or on different days. The types of these two slots are then interchanged without interrupting any other slots, i.e., $X_{ds_k q_1} = 0 \wedge X_{ds_k q_2} = 1$ and $X_{ds_l q_2} = 0 \wedge X_{ds_l q_1} = 1$. An exception to this rule is cases where two slots of different durations are selected. If slot s_l has a longer duration than slot s_k , multiple subsequent slots of the same type as slot s_k are interchanged. In this way, the session duration of each day remains constant. If this is not feasible, new slots are selected. Figure 3 illustrates the interchange operator.

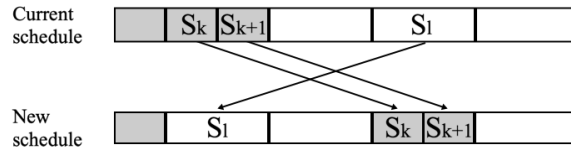


Fig. 3: Illustration of the interchange operator

- The *shift* operator can be operated in either a forward direction or a backward direction, each with a 50% probability. A shift takes place only within a randomly selected day. In a forward shift, a slot s_k is selected and moved forward to an earlier time slot s_l (with $l < k$), i.e., $X_{ds_l q} = X_{ds_k q}$ ($\forall q \in Q$). All slots between time slots s_l and s_k are shifted to one position later, i.e., $X_{d(s+1)q} = X_{dsq}$ ($\forall q \in Q, s = \{s_l, \dots, s_k - 1\}$). Other time slots outside the range of time slots $[s_l, s_k]$ are unchanged.

Figure 4(a) gives an example of a forward shift. In a backward shift, a slot s_k is moved backward to a later time slot s_l (with $k < l$), i.e., $X_{ds_lq} = X_{ds_kq}$ ($\forall q \in Q$). All slots in between these slots are shifted one position earlier, i.e., $X_{dsq} = X_{d(s+1)q}$ ($\forall q \in Q, s = \{s_k, \dots, s_l - 1\}$). Other time slots outside the range of time slots $[s_k, s_l]$ are unchanged. This operator is illustrated in Figure 4(b).

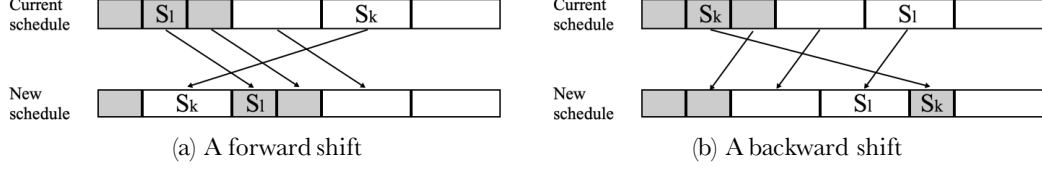


Fig. 4: Illustration of the shift operator

4.1.2 Update of the current schedule and the archive

The replacement of the *current* schedule with another schedule, possibly the *new* schedule or a schedule from the archive, is guided by a certain probability of acceptance Pr_{acc} and (non-)dominance between the two schedules (Bandyopadhyay et al., 2008). Algorithm 3 displays how the *current* schedule and the archive are updated.

(Non-)dominance between schedules - Schedule A dominates schedule B if the average objective function value of schedule A is smaller than or equal to the average objective of schedule B for every objective and is strictly smaller for at least one objective, i.e.,

$Dom(A, B)$

1, if $\forall m : \overline{OV}_m(A) \leq \overline{OV}_m(B)$ and $\exists m : \overline{OV}_m(A) < \overline{OV}_m(B)$

0, otherwise

If $Dom(A, B) = 0$ and $Dom(B, A) = 0$, the two schedules are denoted as non-dominating schedules.

Level of domination - The level of domination between schedule A and schedule B is an aggregate measure reflecting the extent to which one schedule outperforms the other in terms of their objective function values and is calculated as follows:

$$\Omega_{A,B} = \prod_{m, \overline{OV}_m(A) \neq \overline{OV}_m(B)} \left(\frac{|\overline{OV}_m(A) - \overline{OV}_m(B)|}{R_m} \right)$$

In this equation, index m refers to the two different objectives ($m \in \{el, ur\}$) and R_m indicates the range of objective m . This range is calculated as $R_m = \max_{g \in \bar{G}} (\overline{OV}_m(g)) - \min_{g \in \bar{G}} (\overline{OV}_m(g))$, where $\bar{G}$ indicates a set of schedules containing the *current* schedule, the newly constructed schedule and the set of schedules in the archive G_a . The weight of the level of domination compared to the temperature in the probability of acceptance Pr_{acc} is w^Ω . This weight remains constant while the temperature decreases, such that the relative importance of dominance increases as the number of evaluated schedules increases. Note that the level of domination between a schedule and a set of schedules G can be determined by taking the average, i.e., $\Omega_{A,G} = \frac{\sum_{g \in G} \Omega_{A,g}}{|G|}$.

Schedule acceptance - If the *new* schedule is not dominated, i.e., neither by any schedule in the archive nor by the *current* schedule, the *current* schedule is replaced with the *new* schedule and the archive is updated. In that case, the probability of acceptance Pr_{acc} is equal to 1.

Whenever the *new* schedule is dominated by a schedule in the archive or by the *current* schedule, the *new* schedule can still replace the *current* schedule as follows. If the *new* schedule is dominated by at least one schedule in the archive and dominates the *current* schedule, the probability of acceptance is

determined by the archive schedule g^{min} with the lowest level of domination over the *new* schedule ($g^{min} = \{g \in G_a : \Omega_{g,new} = \min_g \Omega_{g,new}\}$), the weight of the level of domination, and the temperature level. If the *new* schedule is not accepted, the *current* schedule is replaced with the schedule g^{min} . If the *new* schedule does not dominate the *current* schedule, the probability that the *new* schedule will be accepted is determined based on the level of domination between the *new* schedule and the set of schedules G^{Dom} that dominates the *new* schedule, the weight of the level of domination and the temperature level. In any case, the probability that the *new* schedule will be accepted as the *current* schedule decreases as the level of domination between any archive schedule(s) and the *new* schedule increases and as the temperature level decreases.

Update of the archive - The newly constructed schedule is added to the archive if it is not dominated by any of the solutions in the archive, i.e., $\text{Dom}(g, new) = 0, \forall g \in G_a$. The archive schedules that are dominated by the newly constructed schedule are removed from the archive.

Algorithm 3 Update *current* and the archive

```

1: Set  $G = G_a \cup \{current\}$ 
2: Determine  $G^{Dom} = \{g \in G : \text{Dom}(g, new) = 1\}$ 
3: If  $G^{Dom} \neq \emptyset$ 
4:   If  $\text{Dom}(new, current) = 1$ 
5:     Find  $g^{min}$ 
6:     Calculate  $Pr_{acc} = 1 - 1 / \left( 1 + \exp \left( - \frac{\Omega_{g^{min}, new} \times w^\Omega}{T} \right) \right)$ 
7:     Generate  $u \in U[0,1]$ 
8:     If  $u \leq Pr_{acc}$ 
9:       Set  $current = new$ 
10:    Else
11:      Set  $current = \text{Archive schedule } g^{min}$ 
12:  Else
13:    Calculate  $\Omega_{new, G^{Dom}}$ 
14:    Set  $current = new$  with  $Pr_{acc} = \exp \left( - \frac{\Omega_{new, G^{Dom}} \times w^\Omega}{T} \right)$ 
15: Else
16:   Set  $current = new$ 
17:   Determine  $G^{Rem} = \{g \in G_a : \text{Dom}(new, g) = 1\}$ 
18:   Set  $G_a = (G_a \setminus G^{Rem}) \cup \{new\}$ 

```

4.2 Simulation

To evaluate a new cyclic appointment schedule generated by the SA algorithm, the average appointment waiting times of elective and urgent patients must be calculated. Appointment waiting times result from operational decisions, and the associated variability is taken into account via simulation. To that end, the operational execution of the appointment schedule is simulated and the arrivals, no-show behaviour, punctuality and service times are simulated. The associated distributions are case-specific and need to be estimated based on historical data. Elective patients are assigned to specific time slots using an online appointment scheduling rule as described in Section 4.2.1. The conditions required to obtain reliable estimates via simulation are discussed in Section 4.2.2.

4.2.1 Patient appointment scheduling

To calculate the waiting times of a patient schedule constructed according to a cyclic appointment schedule, patients have to be scheduled one-by-one in a FCFS order. The patients are assigned appointment times (A_i) according to an appointment scheduling rule. In this study, we examine the impact of variations in the following appointment rules:

– **Rule 1 - FCFS rule**

The appointment time is equal to the start time of the assigned time slot ($A_i = O_i$).

– **Rule 2 - Bailey-Welch rule**

The appointment time of the first K patients of the session on day d in period p is equal to the start time of the first time slot ($A_i = f_{pd}^1$). Each subsequent patient receives an appointment time equal to the appointment time of the previous patient plus the previous patient's expected service duration ($\bar{\delta}_{q^*}$). Note that this rule uses the expected service duration rather than the slot length. The slot length is a managerial decision and is frequently equal to the expected service duration, though this is not necessary per se. The rule can be adapted and generalised by adding parameter k_σ times the standard deviation of the expected scan duration σ_q ($A_i = A_{i^*} + \bar{\delta}_{q^*} + k_\sigma \times \sigma_{q^*}$ with q^* , the slot type of patient i^* , who is the patient scheduled before patient i). The parameter k_σ determines the extent to which possible deviations in the scan duration of the previous patient are taken into account. Giving different values to the parameters K and k_σ leads to different variants of the Bailey-Welch rule.

– **Rule 3 - Blocking rule**

A session is split in blocks consisting of multiple time slots. All the patients with assigned time slots that lie within the same block on day d in period p receive appointment times equal to the start time of the block. This rule can be altered by assuming a different number of slots per block (B) ($A_i = f_{pd}^1 + B_i \times l_B$, where B_i indicates the block to which patient i belongs and l_B indicates the length of the blocks (in minutes)). Notably, similar to the numbering of the time slots, the numbering of the blocks is restarted at 0 each day.

– **Rule 4 - Benchmark rule**

Patients are assigned to appointment times that are either earlier or later than the start times of their slots. We examine the following rules:

- (a) each patient receives an appointment time $k_\sigma \times \sigma_q$ earlier than his or her slot time, i.e., $A_i = O_i - k_\sigma \times \sigma_q$ (where q is the slot type of patient i);
- (b) the first patient of a session is scheduled at its slot time ($A_i = O_i$) or the session start time, as no deviation in scan duration could have taken place yet. Each subsequent patient receives an appointment time $k_\sigma \times \sigma_q$ later than the expected end time of the previous patient ($A_i = A_{i^*} + \bar{\delta}_{q^*} + k_\sigma \times \sigma_{q^*}$, where q^* is the slot type of patient i^* , who is the patient scheduled before patient i);
- (c) the first K patients for a day ($s_i < s_K$) arrive earlier than their slot times ($A_i = O_i - (s_K - s_i) \times k_\sigma \times \sigma_q$), patient $i = K$ ($s_i = s_K$) arrives at his slot time ($A_i = O_i$), and the rest ($s_i > s_K$) arrive later than their slot times ($A_i = O_i + h \times (s_i - s_K) \times k_\sigma \times \sigma_q$). s_i denotes the slot number of patient i with slot type q . h is a parameter that determines how much later appointment times of patients ($s_i > s_K$) are than their slot times. The larger h is, the greater the compensation is for possible deviations in the scan duration of previous patients. This rule is also referred to as the dome rule (Creemers et al., 2020).

4.2.2 Simulation conditions

To obtain reliable interval estimators for the patient waiting times, appropriate simulation conditions should be determined with respect to the number of periods P and the number of simulation runs M . Both of these determinants impact the number of patients simulated, which influences the accuracy of the interval estimators. The values of M and P depend on the balance between the preferred level of

accuracy and the computation time. The choice of these simulation conditions further depends upon the nature of the system. The simulation has a non-terminating character, i.e., the length of a simulation run is not fixed by an explicit natural event but should be determined via experimentation. We conduct multiple independent simulation runs M initialised via different seed values. Each run visits multiple periods P , including a warm-up period W . This warm-up period represents the period at the beginning of the simulation where appointment waiting times are not representative for the system and is therefore not taken into account in the calculation of the averages and interval estimators (Robinson & Bhatia, 1995).

5 Computational Experiments

In this section, the computational experiments used to validate the proposed methodology are described. In Section 5.1, we discuss the test design and input parameters. The experiments are performed using real-life data from the MR department of the Sint-Andries hospital located in Tielt (Belgium). In Section 5.2, we discuss the performance of the obtained cyclic appointment schedules using different appointment scheduling rules and give insights into the timing and number of time slots per patient type needed to generate an adequate performance. In Section 5.3, we benchmark the solutions obtained using the proposed solution procedure with other methods. Section 5.4 discusses the value of integrating strategic and tactical decisions. The experiments are carried out using a 1.60 GHz Dual-Core Intel Core i5 processor.

5.1 Test Design

In the following section, we discuss the input problem characteristics related to the planning and scheduling of patients for the examined case study. In addition, we set the relevant values for the parameters characterising the solution methodology.

5.1.1 Input problem characteristics

Input data related to integrated capacity distribution and scheduling

Strategic and tactical decisions - The hospital currently uses a cyclic appointment schedule with 90 elective "short" slots (15 min), 63 elective "long" slots (30 min) and 13 urgent slots (30 min). The slot length is equal to the expected appointment duration of specific scan types. The cyclic appointment schedule has a length of one week and patients can be scheduled Monday through Saturday but not on Sunday. In general, a session of the radiology department of the hospital begins at 5AM and continues until 9PM. However, no patients are scheduled before 5:45AM, except for on Monday, when patients can be scheduled starting at 5:15AM. The hospital imposes certain case-specific restrictions related to the timing of urgent and elective slots due to predefined breaks (e.g., every day between 6:45AM and 7AM) and the reservation of MRI slots for another hospital (e.g., on Tuesdays between 7AM and 2PM, Thursdays between 7AM and 5PM). These time slots are fixed and considered idle or not available. We assume that the scan scheduled after such an idle or unavailable slot starts at the corresponding patient's arrival time if the scan before the idle or unavailable slot is finished. Urgent patients who arrived during a session are treated during the overtime of that session, i.e., after 9PM, if they cannot be scheduled within that session. In addition, some restrictions related to the allocation of the urgent slots are present. No urgent slots can be scheduled over the weekend or on Tuesday morning. Moreover, there should not be any urgent slots before 7AM or after 7PM. On Thursday, only 2 specific slots in the afternoon can be used as urgent slots. In the computational results below, we consider these case-specific constraints, demonstrating the applicability of the developed algorithm to a real-life case study.

However, to demonstrate the generalizability of the presented findings, we also explore the sensitivity of the results by removing these case-specific constraints. In this experiment, an identical level of capacity in the outpatient department is considered for reasons of comparability and consistency with the other input data.

Appointment scheduling decisions - In this study, eight appointment rules are tested (cf. Section 4.2.1). Table 1 contains the employed variants and their parameter settings, which are set based on the literature. For each appointment rule, only a limited set of parameter values are tested as the main goal is to compare the impact of the logic of different appointment rules on the examined waiting times. Currently, the hospital uses an FCFS appointment rule with patients' appointment times equal to the times of the assigned slots. All the examined appointment rules assign appointment times in the order that patients arrive.

r	Rule	Parameters
1	FCFS	-
2	Bailey-Welch (a) (first K at session start time)	$k_\sigma = 0$; $K = 2$
3	Bailey-Welch (b) (first K at session start time)	$k_\sigma = 0.5$; $K = 2$
4	Bailey-Welch (c) (first K at session start time)	$k_\sigma = 0$; $K = 3$
5	Blocking (block size = B)	$B = 2$
6	Benchmark rule (a) (everyone earlier)	$k_\sigma = 0.5$
7	Benchmark rule (b) (everyone later except the first one)	$k_\sigma = 0.5$
8	Benchmark rule (c) (K earlier, the rest later)	$k_\sigma = 0.5$; $K = (\sum_{s \in S_d} \sum_{q \in Q} X_{dsq})/2$; $h = 1$

Table 1: The considered appointment scheduling rules.

Input data related to operational variability

A single simulation run consists of generating a random patient list for P periods that are one week long. To simulate patient arrivals, no-show behaviour, punctuality and scan durations as accurately as possible, we fit the distributions as closely as possible to the empirical data collected from the examined hospital using SPSS and MATLAB. The obtained distributions are described below.

Uncertainty of patient arrivals - No significant difference in any distribution is found between different weeks and days for either patient type except for days on which the front desk is closed, during which (elective) patients cannot call to make appointments. The following considerations are taken into account with respect to elective and urgent patients:

- The distribution of the call times of *elective patients* for MR scans is estimated based on the historical call logs of the hospital. The appointment desk is only open from Monday to Friday. Hence, no elective patients arrive on Saturday or Sunday. When fitting the empirical data, we find that the number of elective arrivals per day follows a gamma distribution with a shape parameter and scale parameter equal to 4.19 and 4.49, respectively. According to the log-likelihood, the gamma distribution is preferable to several other distributions (e.g., the normal, lognormal and the Weibull distributions). The patient arrivals of every weekday follow the same distribution. The distribution of the arrival times throughout each day follows a complex distribution, such that an empirical distribution is preferable to a fitted theoretical distribution. Figure I in Online Appendix A shows the frequency of patient arrivals.
- The average number of *urgent patients* per day is estimated based on historical data. Every day of the week, the number of urgent patients follows a normal distribution with a mean equal to 1.48 patients per day and a standard deviation equal to 0.93. The normal distribution is preferred to the Poisson, exponential and binomial distributions according to the log-likelihood. In line with the literature (cf. Section 3.2.1), the interarrival times of urgent patients follow a negative exponential distribution, as patient arrivals are typically randomly distributed throughout a day. The mean interarrival time

is $\frac{1}{N_{ur,d}} \times 16 \times 60$ minutes, where $N_{ur,d}$ is equal to the number of urgent patients arriving on day d between 5AM and 9PM.

No-show behaviour - In the historical hospital logs, no-show patients are logged separately from the call logs. This no-show log contains only the arrival day and scan type of each no-show patient. The distribution of the number of no-show patients per day is estimated to follow a truncated normal distribution with a mean of 0.433 patients and a standard error of 0.114. When the distribution returns a negative value, a new random value is requested; thus, the distribution is cut off at zero. As no call time data is available for no-show patients, it is assumed that the empirical arrival distribution of the other elective patients can be applied here as well.

Uncertainty of patient punctuality - Based on historical data, the punctuality of elective patients is estimated to follow a normal distribution, which is consistent with the relevant literature, with a mean of -0.00194 days and a standard error of 0.0118 (i.e., on average, patients arrive 3 minutes before their appointment times).

Patient scan duration - The historical data on scan durations shows a significant difference between certain scan types and no significant difference between urgent and elective patient types. All 40 scan types are classified according to 11 scan type groups based on differences in scan duration, differences in body parts examined (e.g., abdomen or spine) and relative demand (see Table I in Online Appendix A). Within each group, no scan type has a mean or variance that is significantly different¹ from those of the others. A different service duration distribution is determined per scan type group using the maximum log-likelihood method. In line with the literature, all the group types are estimated to follow either a normal or a lognormal distribution. As we assume only 2 elective slot types for simplicity, each group should be assigned to one of these slot types. Based on (historical) scan durations, the elective patients are divided into two classes based on whether their scan duration is shorter or longer than 20 minutes; thus, they can be assigned accordingly to the "short" or "long" elective time slots, respectively.

5.1.2 Input solution methodology parameters

To conduct the computational experiments, the SA parameters (cf. Section 4.1) are set based on preliminary experiments. The initial schedule is the schedule currently used in the hospital. The start temperature is set to 100 and the annealing parameter β to 0.8; these values are often used in the literature to decrease the temperature exponentially and have shown to lead to an adequate convergence towards an optimal solution. The weight of the level of domination (w^Ω) in the probability of acceptance is equal to 1000. When exploring the current solution's neighbourhood in a random manner, it is best to set Pr_1 to 0.5 and Pr_2 to 0.7. To set the stopping criteria, i.e., the number of schedules per temperature level (L) and the total number of temperature levels tested N_{epoch} , the convergence of the SA algorithm is assessed. Based on these results, N_{epoch} is set to 10 and L is set to 200 to ensure convergence and keep the CPU time within reasonable limits.

The accuracy of the simulation results (cf. Section 4.2) is reflected by the confidence interval (CI) and dependent upon the standard error, which can be decreased by increasing the number of simulations M and/or by increasing the number of periods $P - W$ per simulation. Based on the running averages and the CI related to the waiting time of both elective and urgent patients (see Figure II in Online Appendix A for the results), P is set to 310 weeks such that our conclusions are based upon patient waiting times of 10 weeks ($P - W$). To determine the number of simulation runs required, we analyse the convergence behaviour and the confidence intervals of the average appointment waiting times (across-replication data) (see Figures III and IV in Online Appendix A for the results). An appropriate value for M is obtained when the results have sufficiently converged and the standard error is small, i.e., when $M =$

¹ According to an independent sample t-test and Levene's test ($\alpha = 0.95$)

120.

Based on these parameters, the simulation takes approximately 40 seconds per cyclic schedule. The estimated standard errors of $\overline{OV}_{el}$ and $\overline{OV}_{ur}$ are approximately 12.74 and 1.51 hours. The widths of the confidence intervals are 4.51 and 0.55 hours, respectively. Using these parameter values, we perform the simulation for all the schedules and all the appointment scheduling rules. Given these settings, the algorithm finishes constructing a Pareto front using a specific appointment schedule rule after approximately 17.5 hours.

5.2 Results

In this section, insights into the impact of the decisions made on each decision level are provided. To derive suitable cyclic appointment schedules, we first analyse the operational decision-making and performance of the different appointment scheduling rules by comparing the obtained estimated Pareto fronts in Section 5.2.1. In Section 5.2.2, we aggregate the cyclic schedules on these Pareto fronts and derive the characteristics of suitable cyclic appointment schedules with respect to timing and number of urgent time slots.

5.2.1 The impact of appointment scheduling rules

To evaluate the impact of the examined appointment scheduling rules on the patient waiting times, we (i) consider the entire yielded Pareto front consisting of schedules with an optimised capacity allocation to elective and urgent patients and (ii) evaluate the performance of schedules lying on the estimated Pareto front with a particular level of capacity allocated to urgent patients to explore the sensitivity of our findings. In our analysis, we focus only on the significant differences among the rules to highlight the best and worst performing rules, unless explicitly stated otherwise.

Impact on the Pareto front with optimised capacity distribution

Figure 5 presents the estimated Pareto fronts obtained via the proposed methodology that rely on a specific appointment scheduling rule. The estimated Pareto fronts map the cyclic appointment schedules according to their performance, i.e., the average elective patient waiting time ($\overline{OV}_{el}$) (y-axis) and the average urgent patient waiting time ($\overline{OV}_{ur}$) (x-axis). The markers, which are coloured according to each appointment rule, indicate the objective values of the relevant cyclic schedules. Figure V in Online Appendix B reveals the performance of the different appointment scheduling rules in terms of the secondary measures, i.e., the elective scan waiting time and daily overtime. Table 2 presents the performance of the different rules in terms of elective appointment times, the waiting time of urgent patients, the scan waiting time of elective patients, and daily overtime, which are averaged for certain urgent capacity settings (S_{ur}) and given for each appointment rule considering the cyclic appointment schedules that lie on the estimated Pareto front. We display the average values and the confidence intervals ($\alpha = 0.95$). Column ' $S_{ur} \in [0, 26]$ ' contains the averages and confidence intervals of all the schedules lying on the estimated Pareto front associated with each appointment scheduling rule.

The results reveal that the applied appointment scheduling rule has a significant impact on the performance of the constructed cyclic appointment schedules. An analysis of the performance of the schedules on the estimated Pareto frontier leads to the following observations:

- (a) From Figure 5, we can observe that the estimated Pareto front associated with variant (c) of the Bailey-Welch rule (with $k_{\sigma} = 0$; $K = 3$) dominates the estimated Pareto front constructed according to any of the other appointment scheduling rules. The benchmark rules exhibit an inferior performance, whereas the FCFS rule delivers an average performance.

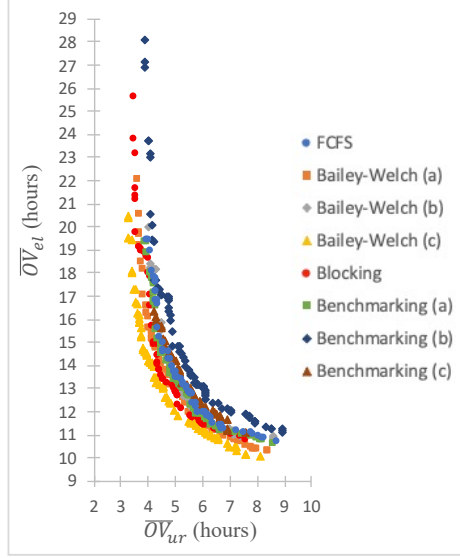


Fig. 5: The performance of schedules lying on the estimated Pareto front regarding the elective and urgent appointment waiting times for each appointment rule

- (b) The appointment scheduling rules have a significant impact on the appointment waiting times of elective patients, which is revealed by the confidence interval estimators ($\alpha = 0.95$) (see Table 2). However, there is no rule that performs significantly better than all the others; thus, we rely on the average as a point estimate to denote the best performing rule. The average values suggest that the variants of the Bailey-Welch rule, particularly variant (b) (with $k_\sigma = 0.5$ and $K = 2$), perform better than the other rules with an average of 13.11 hours. The blocking rule and variant (b) of the benchmark rule both perform significantly worse than all the other rules. Based on the average values, variant (b) of the benchmark rule performs the worst with an average of 15.07 hours.
- (c) Regarding the waiting times of urgent patients, the results reveal that variant (c) of the Bailey-Welch rule (with $k_\sigma = 0$ and $K = 3$) and the blocking rule significantly outperform the other rules, leading to average waiting times of 4.83 and 4.85 hours, respectively. Based on the average values, variant (b) of the benchmark rule performs worse than the other rules with an average waiting time of 5.57 hours for urgent patients.
- (d) However, regarding the elective scan waiting time, the blocking rule and variant (c) of the Bailey-Welch rule exhibit the worst performance with averages of 21.33 and 20.77 minutes, respectively. Variant (c) of the benchmark rule significantly outperforms all the other rules with an average of 16.02 minutes. The benchmark rule variants lead to significantly shorter scan waiting times than the blocking rule and variants (a) and (c) of the Bailey-Welch rule. These results confirm earlier findings from the literature.
- (e) The conclusions regarding daily overtime are approximately the inverse of those on scan waiting times. Variant (c) of the Bailey-Welch rule significantly outperforms all the other appointment rules. This rule leads to an average daily overtime of 9.71 minutes. Variants (b) and (c) of the benchmark rule lead to more than 30 minutes of overtime on average and perform significantly worse than the other appointment rules.

Impact of schedules lying on the Pareto front with a particular urgent capacity setting

Table 2 also presents the performance of the cyclic schedules lying on the estimated Pareto front for certain urgent capacity settings, i.e., ' $S_{ur} = 5$ ', ' $S_{ur} = 10$ ', ' $S_{ur} = 15$ ' and ' $S_{ur} = 20$ ', for the

different appointment scheduling rules to explore the sensitivity of our findings over different capacity distributions. To indicate the relative performance of the appointment scheduling rules, we also display the relative ranking of each rule for each performance measure based on average values. Notably, for some rules, no schedules with 15 or 20 urgent slots were present in the estimated Pareto front.

In general, based on average performance values, we observe that the performance of the blocking rule and especially variant (b) of the benchmark rule are inferior in relation to most of the performance measures whereas the FCFS rule exhibits average performance. The positive impact of the variants of the Bailey-Welch rule, variant (c) in particular, can be clearly discerned. This is supported by the average rankings of the rules. In addition, the results regarding the secondary performance measures show a clear trade-off between scan waiting time and daily overtime for the different appointment scheduling rules. This sensitivity analysis related to the performance of the rules for different capacity settings confirms the earlier findings regarding the entire Pareto front.

Rule	Performance measure	Sur ∈ [0, 26]			Sur = 5			Sur = 10			Sur = 15			Sur = 20			Ranking
		Mean	CI		Mean	CI		Mean	CI		Mean	CI		Mean	CI		
FCFS	OV_{el}	13.69	[13.11, 14.27]		11.52	[11.06, 11.97]		13.44	[12.89, 14]		na	na		19.49	[18.5, 20.48]		5
	OV_{ur}	5.54	[5.3, 5.79]		6.46	[6.2, 6.71]		5.12	[4.87, 5.37]		na	na		3.94	[4.18, 4.18]		3
	H_{el}^{OT}	18.37	[18.05, 18.68]		19.54	[19.19, 19.9]		18.33	[18.02, 18.64]		na	na		15.14	[14.92, 15.36]		5
Bailey-Welch (a)	OV_{el}	25.68	[25.07, 26.29]		30.81	[30.14, 31.47]		23.12	[22.53, 23.71]		na	na		19.77	[19.16, 20.38]		6
	OV_{ur}	13.78	[13.16, 14.39]		11.32	[10.85, 11.8]		13.74	[13.18, 14.29]		16.00	[15.29, 16.7]		19.76	[18.69, 20.84]		6
	H_{el}^{OT}	5.33	[5.09, 5.58]		6.41	[6.16, 6.67]		4.65	[4.41, 4.89]		3.93	[3.69, 4.17]		3.63	[3.4, 3.87]		3
Bailey-Welch (b)	OV_{ur}	19.11	[18.78, 19.44]		19.93	[19.56, 20.29]		19.63	[19.3, 19.96]		17.99	[17.69, 18.29]		16.71	[16.49, 16.94]		6
	H_{el}^{OT}	16.48	[15.98, 16.98]		19.46	[18.95, 19.99]		13.83	[13.35, 14.31]		11.70	[11.24, 12.16]		11.95	[11.49, 12.41]		2
Bailey-Welch (c)	OV_{el}	13.11	[12.56, 13.65]		11.74	[11.29, 12.19]		13.79	[13.19, 14.38]		na	na		na	na		1
	OV_{ur}	5.62	[5.37, 5.86]		6.15	[5.9, 6.39]		4.86	[4.62, 5.1]		na	na		na	na		7
	H_{el}^{OT}	18.42	[18.1, 18.73]		18.89	[18.55, 19.23]		17.74	[17.44, 18.03]		na	na		na	na		5
Blocking	OV_{el}	23.30	[22.72, 23.88]		26.75	[26.16, 27.35]		20.10	[19.53, 20.68]		na	na		na	na		5
	OV_{ur}	13.60	[12.97, 14.23]		11.01	[10.55, 11.46]		12.98	[12.42, 13.54]		15.24	[14.49, 15.99]		na	na		3
	H_{el}^{OT}	4.83	[4.59, 5.08]		6.27	[6.01, 6.53]		4.46	[4.22, 4.71]		3.73	[3.48, 3.97]		na	na		1
Benchmark rule (a)	OV_{ur}	20.77	[20.41, 21.14]		22.05	[21.61, 22.49]		20.91	[20.56, 21.25]		20.15	[19.83, 20.46]		na	na		7
	H_{el}^{OT}	9.71	[9.32, 10.11]		12.54	[12.12, 12.97]		9.47	[9.07, 9.87]		6.85	[6.5, 7.19]		na	na		1
Benchmark rule (b)	OV_{el}	14.93	[14.28, 15.58]		11.74	[11.29, 12.2]		13.50	[12.93, 14.06]		18.37	[17.6, 19.14]		25.66	[24.14, 27.17]		7
	OV_{ur}	4.85	[4.61, 5.1]		5.70	[5.45, 5.96]		4.48	[4.23, 4.73]		3.96	[3.72, 4.2]		3.44	[3.2, 3.68]		2
	H_{el}^{OT}	21.33	[21.02, 21.65]		22.58	[22.22, 22.93]		22.03	[21.73, 22.33]		20.32	[20.02, 20.62]		18.46	[18.22, 18.71]		8
Benchmark rule (c)	OV_{el}	20.75	[20.11, 21.39]		23.98	[23.3, 24.67]		18.49	[17.88, 19.11]		16.05	[15.47, 16.64]		16.30	[15.75, 16.86]		3
	OV_{ur}	13.62	[13.03, 14.2]		11.44	[10.98, 11.9]		13.37	[12.81, 13.93]		na	na		19.42	[18.43, 20.41]		4
	H_{el}^{OT}	5.46	[5.21, 5.71]		6.37	[6.12, 6.63]		5.04	[4.79, 5.28]		na	na		3.86	[3.62, 4.1]		4
Benchmark rule (b)	OV_{ur}	18.38	[18.06, 18.69]		19.57	[19.21, 19.92]		18.33	[18.02, 18.64]		na	na		15.11	[14.89, 15.33]		4
	H_{el}^{OT}	23.03	[22.44, 23.62]		27.76	[27.12, 28.4]		20.68	[20.11, 21.24]		na	na		17.57	[16.99, 18.14]		4
	OV_{el}	15.07	[14.41, 15.73]		12.11	[11.62, 12.6]		13.78	[13.22, 14.33]		17.00	[16.28, 17.72]		na	na		8
Benchmark rule (c)	OV_{ur}	5.75	[5.5, 6]		6.89	[6.64, 7.14]		5.50	[5.25, 5.74]		4.70	[4.46, 4.94]		na	na		8
	H_{el}^{OT}	17.03	[16.75, 17.31]		7.32	[7.03, 7.61]		9.53	[9.24, 9.82]		7.59	[7.29, 7.89]		na	na		2
	OV_{el}	32.46	[31.77, 33.16]		36.96	[36.28, 37.64]		30.66	[29.95, 31.37]		27.74	[27.03, 28.45]		na	na		8
Benchmark rule (c)	OV_{ur}	13.31	[12.76, 13.86]		11.81	[11.36, 12.27]		13.60	[13.05, 14.16]		16.06	[15.34, 16.77]		na	na		2
	H_{el}^{OT}	5.55	[5.3, 5.79]		6.58	[6.32, 6.83]		5.16	[4.92, 5.4]		4.26	[4.02, 4.5]		na	na		6
	OV_{el}	16.02	[15.76, 16.28]		16.89	[16.59, 17.19]		15.90	[15.65, 16.16]		14.93	[14.72, 15.15]		na	na		1
	OV_{ur}	32.32	[31.61, 33.04]		37.73	[37, 38.46]		29.59	[28.87, 30.3]		25.77	[25.09, 26.46]		na	na		7

Table 2: The average performance, confidence interval and ranking of each appointment scheduling rule for all the obtained schedules ($S_{ur} \in [0, 26]$) and for certain isolated schedules with urgent capacity settings of 5, 10, 15 and 20 slots.

5.2.2 The impact of capacity distribution and slot schedules

In this section, we (i) evaluate the impact of the capacity dedicated to elective and urgent patients on patient waiting times and (ii) identify the characteristics of slot schedules that lead to lower waiting times.

Analysis of the Pareto front as a function of capacity distribution

Figure 6 shows the impact of the capacity allocated to urgent patients on the average appointment waiting times. More specifically, we show all the cyclic schedules lying on the estimated Pareto front associated with one of the appointment scheduling rules (cf. Section 5.2.1) and categorise these schedules according to the number of urgent slots. The schedules are mapped according to their performance, i.e., their average elective patient waiting time ($\overline{OV}_{el}$) (y-axis) and average urgent patient waiting time ($\overline{OV}_{ur}$) (x-axis). Colour-coding is used to differentiate between schedules with different numbers of time slots dedicated to urgent patients (S_{ur}). Figure VI in Online Appendix B reveals the performance of the Pareto schedules concerning the secondary performance measures.

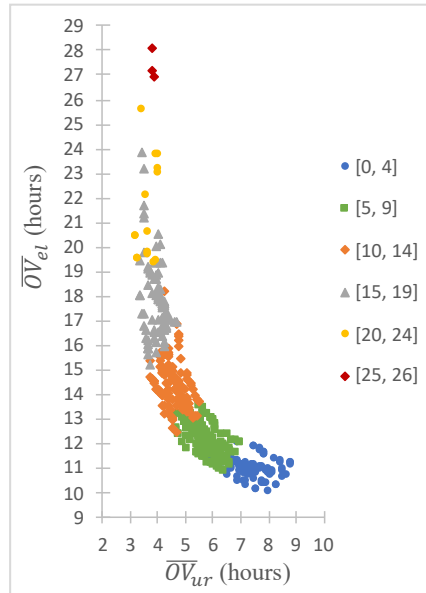


Fig. 6: The impact of urgent capacity on the objective values

The following conclusions can be made concerning the impact of the capacity distribution between elective and urgent patients:

- (a) An increase in urgent capacity has a positive impact on the average urgent appointment waiting time. The detected range of the number of urgent slots is 0 to 26 slots. Beyond 26 urgent time slots, no efficient cyclic schedules are found. On average, the waiting time decreases by 3.15% per added urgent slot. The average urgent appointment waiting time when all the urgent cases are treated during overtime (i.e., $S_{ur} = 0$) ranges between 8.07 and 8.87 hours. When the number of urgent slots is high, e.g., equal to 19, the urgent waiting times range between 3.34 and 4.28 hours. Obviously, the average elective waiting time is negatively impacted by an increase in the number of urgent slots, with an average increase of 3.50% per added slot. The detected range of "short" ("long") elective slots is between 94 and 128 (46 and 62) slots. When there are 100 "short" elective slots, the average elective waiting time for the associated scan types varies from 18 to 21 hours. When there are 122 "short" slots, the average elective waiting time decreases and ranges between only 8 and 10 hours.

- (b) When elective slots are replaced with urgent slots, the average elective scan waiting time slightly decreases. The average range goes from [19,22] minutes to [15,19] minutes when the number of urgent slots is increased from 0 to 19. This is because not all urgent slots are actually filled by the examination of an urgent patient; indeed, if an urgent patient has not arrived, any delays in the patient schedule are mitigated by this idle time slot and the next patient can be examined on time again, provided that the magnitude of the delay is not excessive and the punctuality of the next patient is more or less as expected.
- (c) The average daily overtime decreases significantly when more urgent slots are added, as fewer urgent patients are handled during overtime. When no capacity is allocated to urgent patients, the average overtime of the schedules ranges between 17 and 48 minutes. When 19 slots are allocated to urgent patients, almost no urgent patients are treated during overtime and the average daily overtime ranges between 6 and 26 minutes, which is mostly used to treat delayed elective patients.

Characteristics of suitable cyclic appointment schedules

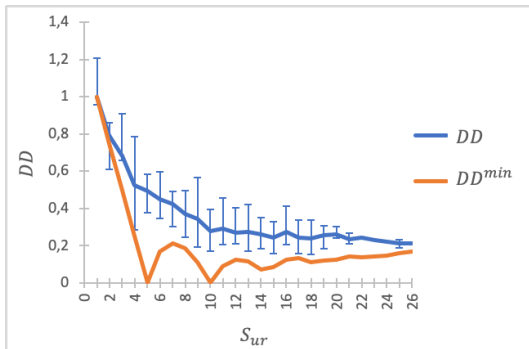
To define appropriate cyclic appointment schedules, we analyse the manner in which the urgent time slots need to be distributed over the planning horizon, given the capacity dedicated to urgent patients. A comparison of the performance of different cyclic schedules reveals that both the number of urgent slots and the timing of urgent slots have a significant impact on the observed appointment waiting times of the constructed cyclic appointment schedules and the resulting Pareto frontier. The impact of both the timing and capacity of urgent slots on the elective scan waiting time, however, is small. The non-dominated cyclic appointment schedules that determine the estimated Pareto frontier demonstrate a good balance with respect to the timing of urgent slots over and throughout the days of the planning horizon. To a limited extent, there is some diversity in the way the time slots for urgent patients are scheduled over the planning horizon depending upon the capacity allocated to urgent patients.

- **Distribution over the days** - To assess the distribution of urgent slots over the examined days, we base our analysis on a general measure of variance proposed by Vanhoucke et al. (2008) for the generation of project networks. This measure, referred to in this paper as the *Day Distribution* (DD), measures the distribution of the time slots dedicated to urgent patients over the examined set of days, and it ranges between 0 and 1. When $DD = 0$, the urgent slots are equally distributed over all days. When $DD = 1$, the number of urgent slots is maximal for one or several days and zero for the other days. Notably, as a result of the different constraints imposed on the schedule, the theoretical smallest value of DD , denoted as DD^{min} , may be different from 0 and is dependent on the number of urgent slots. Figure 7(a) displays the average value and range of the DD -metric (y-axis) as a function of the allocated urgent capacity (x-axis). In addition, we provide the DD^{min} value per urgent capacity setting.

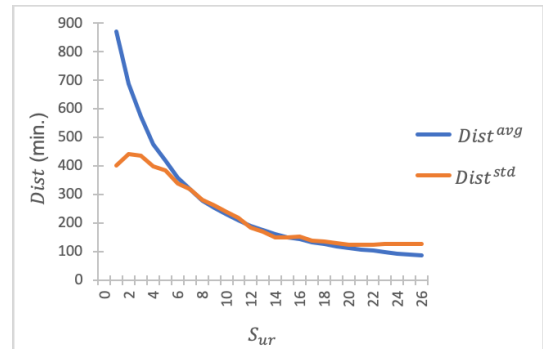
When evaluating the cyclic schedules on the estimated Pareto front, we observe that the DD metric confirms that an equal distribution of urgent slots over the days is preferable. All the schedules tend to have DD -metric values close to DD^{min} , and this is certainly the case when a relatively large amount of capacity is dedicated to urgent patients. Hence, we can state that the capacity reserved for urgent patients is to some extent divided evenly over the days of the cyclic planning horizon, taking the timing constraints for allocating urgent capacity into account. The latter explains why the calculated value of DD is not exactly equal to DD^{min} . When there are few urgent slots, these slots are scheduled primarily towards the beginning of the week. When more capacity is reserved for urgent patients, an equal distribution over the weekdays seems to be preferable, as it facilitates shorter appointment waiting times and improves the balance between Monday, Wednesday and Friday. As fewer time slots are eligible to be used as urgent slots on Tuesday and Thursday, fewer slots are reserved for urgent patients on these days.

– **Distribution within the days** - To assess the distribution of urgent slots within individual days, we employ the distance between consecutive time slots dedicated to urgent patients ($Dist^{avg}$) and the standard deviation of this distance ($Dist^{std}$) over the cyclic scheduling horizon. A lower value of $Dist^{std}$ implies a better distribution of urgent time slots. Notably, in these calculations, the distance between the start of a session and the first urgent time slot and the distance between the last urgent time slot and the end of a session are taken into account. Figure 7(b) displays both $Dist^{avg}$ and $Dist^{std}$ averaged over all the cyclic schedules on the estimated Pareto front per capacity setting. In contrast to the dominated schedules, the following observations are made regarding the cyclic schedules on the estimated Pareto front:

- (a) Within a single day, the urgent slots are distributed evenly to some extent, with more or less equal time intervals between subsequent urgent time slots in line with the reserved capacity for urgent patients. For example, when $S_{ur} = 8$, the average distance between urgent slots ranges between 247 and 282 minutes. The standard deviation of these distances ranges between 167 and 317 minutes, which is rather small, leading to the conclusion that the urgent slots are fairly balanced throughout the days. The average distance between the urgent slots within a day decreases significantly with an increase in the number of urgent slots. In addition to the average, the standard deviation decreases significantly when the urgent capacity increases, which implies that the urgent slots become more balanced throughout the day. When e.g., $S_{ur} = 19$, the average distance between urgent time slots is 120 minutes for all the schedules, and the standard deviation ranges between 123 and 136 minutes. It should be clear that the schedules that group urgent slots together or put them at the end of a session, as is currently done in the hospital, are dominated by more balanced cyclic appointment schedules.
- (b) Urgent slots are scheduled in a rather balanced manner over each day, as the average start time is approximately 1PM, i.e., the midpoint of a session, and this finding is not impacted by the number of urgent slots. Thus, regardless of the number of urgent slots, these slots are not scheduled primarily in the morning or the evening but are equally distributed throughout the day. The time range within which the slots are scheduled is concentrated around 1PM. For example, when $S_{ur} = 8$, slots are typically scheduled between 11AM and 4PM.



(a) The spread over the days (DD) as a function of the urgent capacity



(b) The spread throughout the sessions ($Dist^{avg}$ and $Dist^{std}$) as a function of the urgent capacity

Fig. 7: The timing of urgent slots

5.2.3 A note on the sensitivity of the results

To explore the sensitivity of the results, the same computational experiments are conducted after removing the case-specific constraints that denote the unavailability of time slots as stipulated by the collaborating hospital (cf. Section 5.1.1). A comparison between the results with the case-specific constraints and those without shows that the absolute appointment and scan waiting times are significantly lower without the case-specific constraints. This is obvious, as case-specific constraints often denote the unavailability of time slots during a relatively long time period. During the days after such a period of unavailability, fewer time slots may be idle, as a larger number of patients are waiting to be treated; this leads to longer appointment and scan waiting times. The findings related to the relative performance of appointment scheduling rules, the capacity distribution between elective and urgent patients, and the timing of urgent slots are consistent with the findings presented earlier in this section, which demonstrates the generalizability of the results.

In additional computational experiments, we evaluate the impact of having multiple servers on the characteristics of the constructed cyclic appointment schedules lying on the estimated Pareto front. The results of our experiments with two servers confirm our findings from a single-server setting. In this 2-server setting, the number of urgent slots corresponding to the efficient schedules on the estimated Pareto frontiers range between 11 and 43; this range is 0 to 26 slots with 1 server. This leads to an urgent waiting time decrease of 30%. Average overtime is 75% lower than it is with 1 server. Based on average performance, the best-performing appointment scheduling rule is again variant (c) of the Bailey-Welch rule. In addition, regarding the timing of urgent slots, we observe that it is preferable to equally distribute urgent slots between the two servers and spread them over and throughout the days to minimise patient waiting times.

5.3 Benchmark

In this section, we benchmark the proposed methodology against other methodologies used to schedule urgent patients in outpatient departments. Each methodology is tested considering all the appointment rules. This benchmarking is performed by comparing the incumbent solutions of the following methodologies:

Proposed ($S_{ur} > 0$) - The proposed multi-objective archived simulated annealing heuristic discussed in Section 4. A Pareto front is formed with schedules containing 0 to 26 time slots reserved for urgent patients.

Current ($S_{ur} > 0$) - The current scheduling practices employed in the hospital. This implies that the current cyclic appointment schedule is considered alongside the FCFS appointment scheduling rule. This appointment schedule contains 13 time slots reserved for urgent patients. These time slots are concentrated within the afternoon session of each day.

Equal ($S_{ur} > 0$) - A cyclic appointment schedule containing 1 to 26 time slots reserved for urgent patients is established. These time slots are divided as evenly as possible over and throughout the days of the planning horizon.

Overtime ($S_{ur} = 0$) - No capacity is dedicated to urgent patients, and these patients are treated only during overtime after all elective patients have been provided with medical care. In this way, we minimise the waiting times of elective patients, as all capacity is reserved for elective patients. However, the waiting times of urgent patients are maximised.

Insertion ($S_{ur} = 0$) - No capacity is dedicated to urgent patients but these patients are treated similarly to emergency patients, as they are inserted as soon as possible between elective patients. In

this way, we minimise the waiting times of urgent patients.

Figure 8 compares the performance of the different methodologies. The proposed solution methodology ('Proposed') gives rise to a Pareto front of cyclic appointment schedules with different urgent capacities. The 'Equal' methodology leads to a single cyclic schedule per capacity setting and per appointment rule. The other methodologies, namely, 'Current', 'Overtime' and 'Insertion', lead to a schedule for each appointment rule. The figure displays the performance of these cyclic appointment schedules, i.e., their average elective patient waiting times ($\overline{OV}_{el}$) (y-axis) and urgent patient waiting times ($\overline{OV}_{ur}$) (x-axis). The dots, which are coloured according to the applied methodology, indicate the objective values of the relevant cyclic schedules. Notably, for all the methodologies other than 'Proposed', the capacity distribution and the timing of the different slot types is assumed to be fixed.

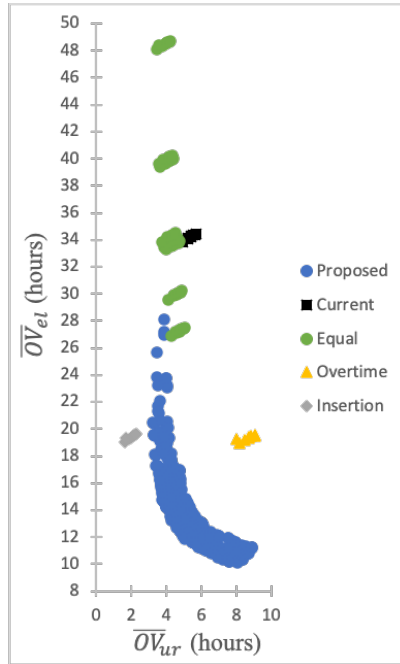


Fig. 8: Benchmark between different methodologies

In the following, we compare the proposed methodology with other methodologies, which is based on our experimental results and discussions with the hospital management. Note that our sensitivity analysis revealed that the findings below are not impacted by the incorporation of case-specific constraints that restrict the timing of elective and/or urgent time slots.

- (a) 'Proposed' *versus* 'Current': The schedule currently utilised in the outpatient department is significantly dominated by different cyclic appointment schedules on the estimated Pareto frontier generated by the proposed methodology. The average waiting times of elective and urgent patients amount to 34.22 hours and 5.37 hours, respectively ('Current').
- (b) 'Proposed' *versus* 'Equal': As mentioned in Section 5.2.2, the proposed methodology produces cyclic schedules very similar to the schedules constructed based on the 'Equal' methodology in terms of the timing of urgent slots. In this way, 55 of the 524 schedules obtained by way of the proposed methodology are dominated by the schedule generated based on the 'Equal' methodology (i.e., with the same rule and capacity allocated to urgent patients) in terms of urgent waiting times. Vice versa,

none of the schedules generated based on the 'Equal' methodology are dominated by the proposed methodology. This can be explained by the fact that the urgent waiting time is the only objective considered in the 'Equal' methodology. However, all the schedules generated based on the 'Equal' methodology are dominated by those constructed with the 'Proposed' methodology in terms of the elective waiting times. As a result, we can conclude that the proposed methodology outperforms the 'Equal' methodology in terms of the following aspects: (i) The proposed methodology takes not only urgent but also elective waiting times into account and enables decision makers to make focused trade-offs between different schedules based on both elective and urgent waiting times. (ii) In contrast to the single-pass heuristic 'Equal', the proposed methodology is capable of dealing with complicated constraints related to the timing of urgent slots.

- (c) 'Proposed' *versus* 'Overtime': When time slots are reserved only for elective patients, the waiting times for these patients are short even if the timing of "short" and "long" elective slots is not optimised. However, treating urgent patients exclusively during overtime leads to very long average waiting times for urgent patients, i.e., 8.62 hours, which is not acceptable for hospital management due to medical reasons amongst others. In addition, the daily overtime observed for the 'Overtime' schedules is significantly higher than that of the other methods, with averages ranging between 21 and 54 minutes.
- (d) 'Proposed' *versus* 'Insertion': Similar to 'Overtime', the 'Insertion' method results in short observed appointment waiting times for elective patients, since $S_{ur} = 0$. In addition, by inserting urgent patients between planned elective patients, the waiting times of urgent patients are minimised as much as possible with the given scheduling constraints, and their average amounts to only 2 hours. However, the hospital management indicated that the scan waiting times for elective patients are very high, i.e., 24 minutes, which is unacceptable for reasons of patient satisfaction and quality of service.

5.4 Value of integrating strategic decisions into cyclic appointment scheduling

The problem under study considers different types of decisions related to (i) the strategic capacity distribution between elective and urgent patients, (ii) tactical decisions related to the timing of slots for elective and urgent patients over a cyclic planning horizon and (iii) operational decisions that determine the appointment times of patients to construct a patient schedule. These decisions are considered simultaneously in the proposed approach. The consideration of operational decisions is indispensable in accurately assessing the operational performance of a cyclic appointment schedule. However, as described in the literature review, strategic decisions are often neglected and the assumption is that decisions are made in a sequential manner. To discern the value of integrating strategic decisions into the construction of a cyclic appointment schedule and to justify the definition of the problem under study, we evaluate the following two methodologies:

- '**Strategic** $\oplus$ **Tactical**': Strategic and tactical decisions are integrated and performed *simultaneously* following the proposed problem description and methodology (cf. Section 3 and 4).
- '**Strategic** $\rightarrow$ **Tactical**': Strategic and tactical decisions are made *sequentially*, i.e., we first fix the capacity allocated to urgent patients and then determine the Pareto front of cyclic appointment schedules according to the capacity setting. To do so, small adaptations to the methodology are required. The *capacity operator* is removed from the heuristic and new initial schedules are created for every relevant capacity setting based on the 'Current schedule', keeping the relative capacity distribution between the elective patient types unchanged as much as possible.

In this experiment, all the relevant fixed urgent capacity settings, which range between 0 and 25 urgent slots, are tested. Figure 9(a) contains some example estimated Pareto frontiers obtained with the 'Strategic $\rightarrow$ Tactical' methodology and the FCFS rule. The findings related to the appointment scheduling rules and the timing of urgent slots provided above are confirmed, i.e., the cyclic schedules on the estimated Pareto front employ best variant (c) of the Bailey-Welch rule, and urgent slots are preferably equally distributed over and throughout the days of the planning horizon to minimise patient waiting times.

Figure 9(b) compares the 'Strategic $\uplus$ Tactical' methodology with the 'Strategic $\rightarrow$ Tactical' methodology. For both methods, the cyclic appointment schedules derived for each appointment rule and urgent capacity setting are assembled and dominated schedules are removed. The figure shows that the estimated Pareto frontier derived via the methodology 'Strategic $\rightarrow$ Tactical' forms a Pareto frontier that is worse than the one derived using the 'Strategic $\uplus$ Tactical' methodology, showing the value of integrating strategic and tactical decisions.

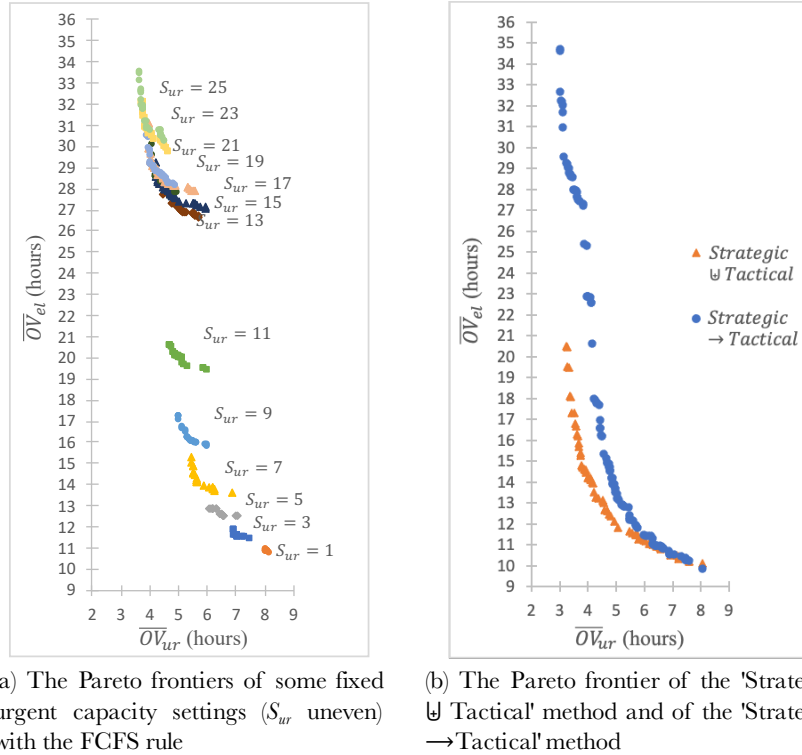


Fig. 9: The value of integrating strategic and tactical decisions

These results can be partly explained by the fact that the integrated methodology 'Strategic $\uplus$ Tactical' offers the flexibility of changing the capacity distribution between different elective patient types. As a result of this increased flexibility, other cyclic appointment schedules with different timing for the slots reserved for urgent patients can be visited in the optimisation search. To validate these observations, we compare the results regarding the different performance measures obtained by the schedules on the estimated Pareto front for the 'Strategic $\uplus$ Tactical' methodology, the 'Strategic $\rightarrow$ Tactical' methodology and the current schedule ('Current') for a fixed capacity setting, i.e., ' $S_{ur} = 13$ '. Based on the results of this comparison, which are displayed in Table 3, the following observations can be made. In contrast to the current schedule ('Current'), the heuristic is capable of significantly improving all the

waiting times using both the integrated methodology and the sequential methodology with fixed capacity distribution. Comparing these latter two methodologies, we observe that the elective appointment waiting times decrease by 44% when strategic decisions and tactical decisions are integrated. This is a result of more efficiently distributing capacity between elective patient types. In addition, the 'Strategic $\oplus$ Tactical' method is capable of identifying schedules that lead to shorter average urgent waiting times with the same number of urgent slots. Because of the larger flexibility for allocating urgent slots, the integrated methodology attains a more even distribution of urgent slots throughout and between the days of the planning horizon, which is confirmed via the lower average values of DD and $Dist^{std}$. Regarding the secondary performance measures, the obtained schedules do not show a significant difference in the elective scan waiting times. Average daily overtime, on the other hand, decreases by 30% when strategic decisions are integrated.

Methodology	$\overline{OV}_{el}$		$\overline{OV}_{ur}$		$\overline{H}_{el}$		$\overline{OT}$	
	Mean	CI	Mean	CI	Mean	CI	Mean	CI
Current	34.22	[32.07, 36.36]	5.37	[5.1, 5.64]	20.20	[19.89, 20.51]	21.09	[20.5, 21.67]
Strategic $\oplus$ Tactical	15.15	[14.49, 15.82]	4.30	[4.06, 4.54]	18.42	[18.13, 18.71]	16.95	[16.4, 17.51]
Strategic $\rightarrow$ Tactical	27.26	[24.3, 30.23]	4.75	[4.5, 5]	18.07	[17.81, 18.34]	24.15	[23.56, 24.73]

Table 3: Results regarding the primary and secondary performance measures for the schedules with $S_{ur} = 13$ per methodology.

6 Conclusions

In this research, we study the cyclic appointment system of a real-life outpatient department. More specifically, we examine the capacity distribution between elective and urgent patients and the scheduling of urgent time slots over a planning horizon to optimise the appointment waiting times of both elective patients and urgent patients. To accurately measure the corresponding performance with respect to patient waiting times, operational decisions related to the online scheduling of individual patients are taken into account. In this way, in contrast to most relevant studies in the literature, we integrate strategic, tactical and operational decisions and focus not only on elective patients but also on urgent patients. We propose a solution methodology that computes a Pareto front of cyclic appointment schedules based on the patient waiting times of elective and urgent patients. This Pareto front is formed using a multi-objective archived simulated annealing heuristic that relies on a schedule archive and exploits different dedicated operators to optimise the problem under study. The operational performance of the resulting cyclic appointment schedules, which are generated by the optimisation algorithm, is evaluated using a simulation, imitating the arrival patterns, punctuality, no-show behaviour and scan durations of individual patients. This evaluation based on a simulation guides our optimisation search process towards better schedules.

In our computational experiments, we discuss the impact of appointment scheduling rules, the timing of urgent time slots and the capacity distribution between elective and urgent patients. This research extends previous results from the literature, which only consider a fixed capacity distribution between different types of patients and evaluate the impact of online scheduling rules only for elective patients. From the results, we conclude that all three types of decisions have a significant impact on the objective values. The appointment rule significantly impacts not only appointment waiting times but also elective scan waiting times and daily overtime. The Bailey-Welch rule has a positive impact on appointment waiting times and overtime, while the benchmark rule returns the lowest elective scan waiting times. The cyclic schedules on the estimated Pareto front of each rule are characterised by a relatively equal distribution of urgent slots over and throughout the days of the planning horizon. The benefits of the

proposed methodology are validated by a benchmark with relevant single-pass solution methodologies. In addition, we highlight the value of integrating strategic and tactical decisions to construct an appropriate cyclic appointment schedule.

There are different ways to extend this study through future research. First, in this paper, the composition of the cyclic appointment schedule is based on scan types requiring shorter or longer time slots. However, multiple specific scan types may be considered based on the examined body part to compose a cyclic appointment schedule accounting for the setup times needed for different scan types. A second interesting extension relating to the problem definition would be to relate the composition of a cyclic appointment schedule to strategic case-mix planning decisions, which deal with choosing the ideal composition and volume of patients to optimise a hospital's financial performance, and accurately determine the required resources based on operational performance. Finally, following the framework of a simulation-based optimisation approach, the operational performance of a schedule is evaluated after it is constructed, which requires a significant amount of computational effort given the large number of schedules generated. More efficient approaches may consider uncertainty proactively and rely on a stochastic optimisation approach such as Sample Average Approximation to build a cyclic appointment schedule that minimises the expected waiting times of elective and urgent patients.

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